

Digital Sentiment Analysis for Quantitative Bitcoin Trading

Methods, Implementation, and Performance Evaluation

Bachelor thesis

Client:	FHNW School of Business
Author:	Leo David Hubmann
Lecturer:	Dr. Pascal Moriggl
City, Date:	Solothurn, 07.08.2025

Digital Sentiment Analysis for Quantitative Bitcoin Trading

Methods, Implementation and Performance Evaluation

Author

Leo Hubmann

leo.hubmann@students.fhnw.ch

Lecturer

Dr. Pascal Moriggl

University of Applied Sciences and Arts

Northwestern Switzerland

pascal.moriggl@fhnw.ch

Solothurn, August 2025

Declaration of Authenticity

I hereby declare

- that I have written this performance record independently and on my own;
- that I have correctly cited all text passages not originating from myself according to common scientific citation rules and that I have clearly mentioned the sources used;
- that I have declared in an index all aids used (e.g. AI tools such as chatbots [e.g. ChatGPT], translation/paraphrasing tools) or programming applications [e.g. Github Copilot] and have indicated their use at the corresponding text passages;
- that I have acquired all intangible rights to any materials I may have used, such as images or graphics, or that these materials were created by me;
- that the topic, the paper or parts of it have not been used in a performance record of another module, unless this has been expressly agreed with the lecturer in advance and is indicated in the paper;
- that I am aware that my work may be checked for plagiarism and for third-party authorship of human or technical origin (artificial intelligence);
- that I am aware that the School of Business FHNW will pursue a violation of this declaration of authenticity or its underlying obligations of the study and examination regulations of the School of Business FHNW and that disciplinary consequences (reprimand or expulsion from the study program) may result.

Place, date

Solothurn, 07.08.2025

Signature

A handwritten signature in black ink, appearing to be 'L. H. H.', followed by a horizontal line.

Abstract

The Bitcoin market is characterised by extreme volatility and a high sensitivity to investor sentiment, posing significant challenges for traditional quantitative trading approaches. This thesis explores, develops, and evaluates a sentiment driven trading strategy designed to navigate this complex environment by systematically exploiting information inefficiencies.

Using publicly available data from Reddit spanning 2021 to 2024, the study employs a dual sentiment analysis approach, combining a lexicon-based method (VADER) with a transformer based machine learning model (FinBERT). These are synthesized into a robust composite sentiment indicator, which drives a rule-based tactical allocation strategy that rebalances weekly according to algorithmically defined market regimes. The prototype is evaluated through backtesting with a strict out-of-sample period covering all of 2024.

The results demonstrate a significant performance improvement over passive benchmarks. In the out-of-sample period, the sentiment driven strategy achieved a total return of 143% against 121% for a buy-and-hold approach, while delivering a substantially higher Sharpe ratio of 2.70 versus 1.49. Crucially, the strategy proved its value in risk management by halving the maximum drawdown from 26% to 14%. The composite sentiment signal was shown to be more effective than either of its basic parts, and a volatility-aware variant of the strategy further enhanced downside protection.

This research provides empirical evidence that challenges the weak-form Efficient Market Hypothesis in the context of Bitcoin and supports the beliefs of behavioural finance. It delivers a transparent, replicable, and accessible framework that proves digital sentiment can be a valuable factor for achieving superior risk-adjusted returns and effectively managing the intrinsic volatility of cryptocurrency markets.

Acknowledgment

First and foremost, I would like to express my gratitude to my supervisor, Dr. Pascal Moriggl, for his openness to my topic ideas and his guidance throughout the process of writing my first thesis.

I am also grateful to those who contributed their time and thoughts during the ideation and writing phase. Special thanks to Marco Hafid for helping me discover possible topics for a bachelor thesis, Christos Antoniou for his insights on efficient interference, and Konstantinos Dimitrakopoulos for recommending the use of dynamic thresholds in sentiment based trading.

My sincere thanks go to the lecturers of my major in Business Analytics. I would like to particularly acknowledge Prof. Dr. Rainer Telesko for introducing me to Python, which laid the foundation for much of my technical work. I am also especially grateful to Dr. Gwendoline Wilke, whose introduction to the tidy framework and valuable resources on Python, R, and time series analysis greatly supported my research. Furthermore, I would like to thank Dr. Stephan Jüngling for introducing me to Google Colab, VADER, and BERT, which became essential tools in my analysis.

Finally, I would like to thank my parents for attempting to read my bachelor thesis - emphasis on attempting.

Table of Contents

Abstract	III
Acknowledgment	IV
1 Introduction	1
1.1 Problem Statement	2
1.2 Research Questions	3
1.3 Delimitations	4
1.4 Approach	5
2 Literature Review	6
2.1 Theoretical Foundations: Market Efficiency, Behavioural Finance, and Bitcoin	6
2.1.1 Efficient Market Hypothesis (EMH)	6
2.1.2 Limitations of EMH in traditional Financial Markets	6
2.1.3 Behavioural Finance	7
2.1.4 Bitcoin as an Asset and Market	7
2.2 Sentiment Analysis in Financial and Cryptocurrency Markets	9
2.3 Data Sources	10
2.3.1 Publicly Available Sources	10
2.3.2 Paid Data Sources	11
2.4 Sentiment Analysis Techniques for Bitcoin Trading	11
2.4.1 Lexicon Based and NLP Methods	11
2.4.2 Machine Learning Based Methods	12
2.4.3 Comparative Analysis and Practical Suitability	13
2.5 Bitcoin Trading Strategies	14
2.5.1 Quantitative Trading Strategies	14
2.5.2 Machine learning based strategies	15
2.5.3 Sentiment Driven Trading Strategies	16
3 Methodology	17

3.1	Research approach	17
3.2	Research Tools and Techniques.....	18
3.2.1	Sentiment Analysis Tools.....	18
3.2.2	Data Collection and Sources	18
3.2.3	Data Preprocessing & Feature Engineering.....	20
3.2.4	Trading Strategy Development.....	20
3.2.5	Strategy Backtesting and Evaluation.....	20
4	Prototype development	22
4.1	Data Processing & Feature Engineering	22
4.1.1	Data Collection.....	22
4.1.2	Text Data Preprocessing.....	24
4.1.3	Sentiment Analysis: VADER.....	25
4.1.4	Sentiment Analysis: FinBERT	26
4.2	Exploratory Data Analysis: Sentiment and Price.....	28
4.2.1	Bitcoin Price Trends.....	28
4.2.2	Sentiment Series Overview	30
4.2.3	Sentiment vs. Price	35
4.3	Strategy	42
4.3.1	Strategy Design Framework.....	42
4.3.2	Signal-engineering methodology.....	43
4.3.3	Parameter Selection & Validation	43
4.3.4	Strategy Variants	44
4.3.5	Implementation Considerations.....	45
5	Backtesting & Results	47
5.1	Backtesting Methodology	47
5.1.1	Temporal Structure and Data Splits	47
5.1.2	Execution Assumptions	48

5.1.3	Benchmark Strategies	48
5.1.4	Performance Metrics	48
5.1.5	Parameter Selection Process.....	49
5.2	Empirical Results.....	50
5.2.1	In-Sample Performance (2021-2023)	50
5.2.2	Out-of-Sample Performance (2024)	50
5.2.3	Extended Benchmark Comparison	52
5.3	Risk Analysis	53
5.3.1	Drawdown Analysis	53
5.3.2	Allocation Dynamics.....	54
5.3.3	Correlation Analysis.....	55
5.4	Robustness & Sensitivity Analysis	56
5.4.1	Comparative Analysis of Sentiment Signals	56
5.4.2	Implementation Robustness.....	57
5.5	Discussion of Results.....	58
6	Conclusion & Future Work.....	60
6.1	Summary of Findings.....	60
6.2	Future Work.....	61
6.3	Final Remarks	61
	References	62
	List of Figures	67
	List of Tables.....	69
	List of Abbreviations.....	70
	Appendix	71

1 Introduction

In recent years cryptocurrency markets, particularly Bitcoin, have attracted significant attention from investors both retail and professional traders, and researches, largely due to their volatility, speculative nature, and sensitivity to investor emotions (Mai et al., 2018). With Bitcoin surging to new heights in mid 2025 and testing price levels above \$117,000, the market has demonstrated continued momentum following the Bitcoin halving in April 2024 and the U.S elections. (Coinmarketcap 2025). The price dynamics of Bitcoin frequently correlate with sentiment expressed on social media platforms, news coverage, and investor behaviours. Studies indicated that public sentiment indicators , such as posts on X (formerly Twitter) and Google search volume, can serve as leading signals for Bitcoin price movements (Bollen et al., 2011; Kristoufek, 2013).

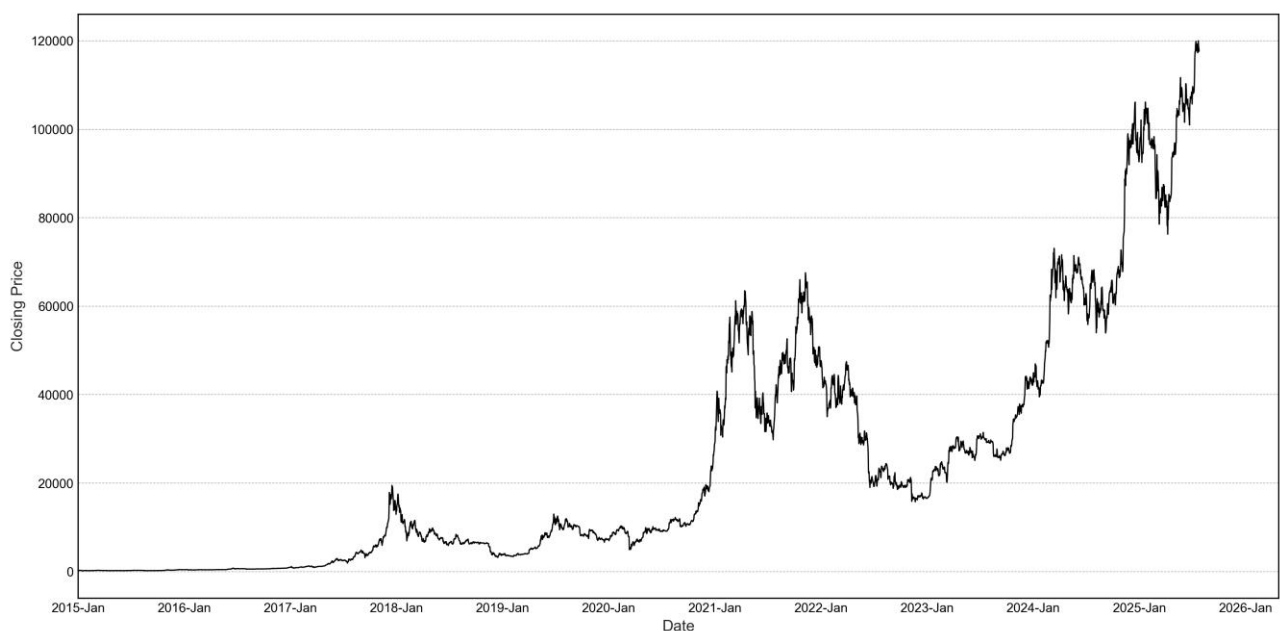


Figure 1: Bitcoin Daily Closing Prices (until March 2025)

Quantitative trading broadly refers to the systematic application of statistical, mathematical, or computational methods to inform trading decisions (Ernest P. Chan, 2021). Sentiment driven trading strategies represent a specialized form of quantitative trading, leveraging digital sentiment signals derived from textual data through Natural Language Processing (NLP) or machine learning (ML) techniques. Given that Bitcoin is responsive to market sentiment and speculative investor behaviour, sentiment analysis seems to be particularly suited to inform quantitative trading decisions in this context.

This thesis explores sentiment analysis in Bitcoin, specifically focusing on the extraction, quantification, and exploratory application of sentiment analysis within a quantitative strategy. By comparatively evaluating sentiment analysis methods and investigating their integration into a prototype trading strategy, this research aims to identify meaningful relationships and assess the feasibility and potential value of a sentiment driven

approach. The goal is to enhance understanding of sentiment's role in Bitcoin market dynamics rather than definitively optimizing strategy performance or implementing a live strategy. The following sections outline the specific problem addressed by this research, the key research questions, and the scope of the investigation.

1.1 Problem Statement

Bitcoin is characterized by extreme volatility, speculative investor behaviour, and significant price movements, often occurring without clear fundamental drivers (Baur et al., 2018; Kyriazis et al., 2020). Quantitative trading approaches typically rely on historical price patterns and technical indicators, but cryptocurrency markets present unique challenges due to their sensitivity to sentiment and investor emotions.

Research by Almeida and Gonçalves (2023) demonstrates that investor sentiment, herd behaviour, and news significantly impact cryptocurrency prices, with investor attention being a key factor driving decisions in these markets. This suggests that incorporating sentiment analysis into trading strategies could potentially enhance their effectiveness, particularly for capturing short term market dynamics in Bitcoin trading.

However, effectively applying sentiment analysis to Bitcoin trading presents multiple challenges. These include accurately extracting sentiment from noisy textual data, quantifying the sentiment meaningfully, and practically integrating these signals into actionable trading strategies (Kraaijeveld & De Smedt, 2020). Additionally, uncertainty remains regarding which sentiment extraction methods perform best for cryptocurrency markets, how sentiment signals influence trading performance in practice, and whether sentiment based approaches actually contribute to superior alpha generation (Garcia & Schweitzer, 2015; Vidal-Tomás et al., 2019).

Despite the growing interest in sentiment analysis for financial markets, a systematic exploration of sentiment driven Bitcoin trading strategies remains limited. Existing research has primarily focused on establishing correlations between sentiment and price movements rather than developing and testing practical trading implementations.

This thesis addresses these gaps by conducting an exploratory analysis of a sentiment driven quantitative Bitcoin trading strategy. It specifically examines the reliability of digital sentiment extraction, quantification methods, and practical integration of sentiment signals into trading decisions. Through this exploratory approach, the research aims to clarify the potential value, feasibility, and practical limitations of sentiment analysis in developing a quantitative trading strategy for Bitcoin markets and comparing it to traditional strategies.

1.2 Research Questions

Building upon the problem statement, this thesis explores sentiment driven quantitative Bitcoin trading strategy through the following research questions:

RQ1: *How can digital sentiment from various online sources be reliably extracted, quantified, and integrated into Bitcoin trading strategies?*

- **Objective:** Define sentiment in a financial context, explore publicly available and paid data sources, and review sentiment extraction methods.

RQ2: *Which sentiment analysis techniques are most effective for creating sentiment feeds to inform Bitcoin trading decisions?*

- **Objective:** Compare and evaluate sentiment analysis methods, such as lexicon based and machine learning based, determining their practical effectiveness, suitability and feasibility within an exploratory context.

RQ3: *How does the inclusion of sentiment data influence alpha generation and overall performance of Bitcoin trading strategies, and which factors significantly drive these results?*

- **Objective:** Assess if a quantitative sentiment driven strategy generates alpha compared to strategies without sentiment indicators and analyse key performance metrics.

Through these research questions, the thesis aims primarily to explore, clarify, and evaluate the potential of digital sentiment analysis as a practical method for creating a quantitative Bitcoin trading strategy.

1.3 Delimitations

This thesis operates within defined boundaries to ensure feasibility and maintain focus on the core research questions.

- **Asset Focus:** The research exclusively examines Bitcoin (BTC) rather than cryptocurrencies or traditional financial assets. This focus allows for deeper analysis of sentiment dynamics in the most established cryptocurrency market.
- **Time Period:** Due to practical constraints in data access, availability and financial resources, this thesis focuses exclusively on historical data from 2021 to 2024 for sentiment analysis and strategy benchmarking.
- **Sentiment Sources:** The analysis primarily utilizes publicly available sentiment data from selected sources (Reddit, EODHD, Google search data) rather than comprehensive coverage of all potential sentiment sources or expensive proprietary paid data services.
- **Methodological Approach:** This research has an exploratory scope, aiming to identify potential relationships and assess feasibility rather than implementing a live trading strategy or proving causality.
- **Implementation Scope:** The thesis develops a prototype trading strategy for back testing purposes only, without implementing a live trading system or addressing the operational challenges of real-time deployment.
- **Performance Metrics:** While the research evaluates strategy performance against baseline approaches (e.g., buy-and-hold, DCA), it does not claim to develop an optimal trading strategy or maximize specific performance metrics.
- **Sentiment Analysis Techniques:** The research focuses on comparing selected sentiment analysis methods (FinBERT and VADER) rather than comprehensively evaluating all available techniques.

The described delimitations ensure that the research remains focused and feasible within the scope of the thesis while still providing valuable insights into the potential of sentiment analysis for Bitcoin trading strategies.

1.4 Approach

This research utilizes an exploratory approach to investigate the potential of sentiment analysis for Bitcoin trading strategies. The methodology consists of several interconnected phases:

First, a thorough literature review establishes the theoretical foundation, examining market efficiency theories, behavioural finance principles, and Bitcoin's unique market characteristics. This review also explores sentiment analysis techniques and their applications in financial markets.

Second, the research develops a methodological framework for extracting and quantifying sentiment from digital sources. This involves selecting appropriate sentiment analysis techniques (FinBERT and VADER) based on their suitability for cryptocurrency-related content and implementing them using Python's natural language processing libraries.

Third, historical Bitcoin price data (2021-2024) is collected alongside sentiment data from selected public sources. Both datasets undergo preprocessing to ensure quality and compatibility, including cleaning, normalization, and alignment of time series data.

Fourth, a prototype trading strategy incorporating sentiment signals is developed. This strategy translates sentiment indicators into trading decisions using predefined rules and parameters.

Fifth, the sentiment based strategy is backtested against historical data and compared with benchmark approaches. Performance evaluation focuses on risk-adjusted metrics including Sharpe ratio and maximum draw-down.

Finally, the results are analysed to assess the potential value of sentiment analysis in Bitcoin trading, identify key factors influencing performance, and determine the practical limitations of the approach. This analysis directly addresses the research questions while acknowledging the exploratory nature of the study.

2 Literature Review

2.1 Theoretical Foundations: Market Efficiency, Behavioural Finance, and Bitcoin

2.1.1 *Efficient Market Hypothesis (EMH)*

In 1970 Eugene Fama defined efficient markets as those in which stock prices fully and instantly reflect all available information. In his theory, Fama outlined three distinct forms of market efficiency: weak, semi-strong, and strong.

Fama's weak form efficiency defines that current asset prices reflect all historical market information, including past prices and trading volumes. In consequence, technical analysis, which relies on historical price patterns cannot consistently yield above-average returns (Fama, 1970).

The semi-strong form suggests that asset prices incorporate all publicly available information, such as financial statements and news. As a result, fundamental analysis, which involves studying the financial health of a company, is unlikely to provide a constant advantage in predicting future prices. On top of this, the strong form expands even further and states that private information is reflected in asset prices. This implies that even insider information is insufficient in providing excess returns (Fama, 1970; Troy Segal, 2024).

2.1.2 *Limitations of EMH in traditional Financial Markets*

The Efficient Market Hypothesis suggests that predicting future asset prices is impossible since today's prices already reflect all available information. However, this theory has been increasingly challenged by empirical evidence suggesting that markets are not always perfectly rational. For example, persistent market anomalies such as momentum trading, mean reversion, and seasonal effects like the January effect contradict the assumption that markets fully incorporate all information into prices (Alajbeg et al., 2012).

Additionally, Robert Shiller's (2003) study on excess volatility showed that asset prices frequently experience fluctuations greater than those justified by changes in their fundamental values. Such findings suggest that psychological factors and irrational behaviours may play a significant role in driving market movements. These limitations highlight gaps in traditional financial theories and open the door for alternative approaches to understanding market dynamics (Shiller, 2003).

One such approach is behavioural finance, which challenges EMH by examining how cognitive biases and emotional influences affect investor decision-making. This perspective will be explored further in the next section.

2.1.3 Behavioural Finance

Behavioural Finance challenges traditional economic assumptions of rational decision making. At its core, this field examines how psychological factors and cognitive biases influence investor behaviour, resulting in market anomalies and price inefficiencies that traditional models like the EMH cannot fully explain (Barberis & Thaler, 2002; Shiller, 2003).

Central concepts in behavioural finance include loss aversion, where investors show greater sensitivity to losses than gains of equal value, leading to asymmetric risk behaviour (Kahneman & Tversky, 1979). Another critical aspect is herding behaviour, where investors follow the actions of others, often ignoring their private information or analysis, resulting in amplified market trends and potential mispricing of assets (Bikhchandani & Sharma, 2000).

In cryptocurrency markets, particularly Bitcoin, behavioural biases are boosted due to high market volatility, significant retail investor presence, and speculative investment behaviour (Almeida & Gonçalves, 2023). Social media platforms further intensify these biases, creating echo chambers and amplifying herd behaviour, emotional trading, and speculative bubbles (Garcia & Schweitzer, 2015).

Understanding these behavioural dynamics is essential for assessing the effectiveness of sentiment based trading strategies, as investor sentiment directly influences short-term market movements and potential trading opportunities in Bitcoin (Kraaijeveld & De Smedt, 2020; Mai et al., 2018).

Therefore, behavioural finance provides a valuable theoretical framework for understanding market anomalies that traditional EMH cannot effectively explain, particularly in the context of cryptocurrency markets.

For quantitative traders, understanding the theoretical frameworks of EMH and behavioural finance is crucial. EMH provides a baseline assumption of market rationality, indicating limited opportunities for excess returns. In contrast, behavioural finance identifies psychological biases and emotional factors that generate exploitable inefficiencies. Given this thesis's focus on sentiment driven Bitcoin trading strategies, these theoretical perspectives are particularly relevant: sentiment analysis explicitly targets market inefficiencies resulting from investor biases, potentially enabling alpha generation. Hence, these frameworks help clarify the foundations behind integrating digital sentiment into quantitative trading models, directly informing the thesis's research questions on sentiment extraction, sentiment analysis, and performance evaluation.

2.1.4 Bitcoin as an Asset and Market

Bitcoin appeared in 2009 as the first decentralized cryptocurrency, characterized by a digital ledger technology known as blockchain, allowing peer-to-peer transactions without financial intermediaries (Nakamoto, 2008).

Initially created as a peer-to-peer electronic cash system, Bitcoin's role has evolved significantly. Today Bitcoin is mainly treated as speculative asset or digital commodity rather than an actual currency. This due to

its substantial price volatility, limited use as a medium of exchange, and speculative investor motivations (Baur et al., 2018; Yermack, 2014).

Empirical research further supports this perspective, demonstrating that users primarily acquire Bitcoin as a speculative investment rather than for transactional purposes. Glaser et al. (2014) found that new users entering the Bitcoin ecosystem mainly engage in speculative trading, which is evident from increased activity on exchanges without corresponding growth in blockchain-based transaction volumes. This behaviour suggests that Bitcoin's attraction largely lies in its potential as an alternative investment vehicle rather than a functional currency.

Bitcoin's asset-like characteristics, including scarcity, limited supply, and lack of intrinsic value, attract investors seeking portfolio diversification, store-of-value opportunities, or speculative gains. (Baur et al., 2018; Yermack, 2014) A systematic literature review examining investor behaviour in cryptocurrency markets by Almeida & Gonçalves (2023) further underscores distinct characteristics in this market. These include irrational investor behaviours predominantly driven by sentiment, uncertainty regarding fundamental value, and high speculative trading volumes. Such traits increase Bitcoin's responsiveness to market sentiment and emotional investment decision making.

Bitcoin's market structure differs from traditional financial markets, which typically have fixed opening hours. The new markets for cryptocurrencies are open 24/7 across fragmented global exchanges with varying liquidity profiles. (Brandvold et al., 2015) Recent research and surveys by Fang et al. (2022) notes increasing market maturity with growing institutional participation, such as asset managers, alongside retail investors, who have historically dominated the market. Institutional involvement contributes to evolving dynamics that increasingly influence Bitcoin's price behaviour and market characteristics.

Regarding market efficiency, Corbet et al. (2019) found that Bitcoin markets initially showed significant inefficiencies but developed toward greater efficiency over time. Bitcoin markets have progressively evolved towards greater informational efficiency but anomalies during episodes of heightened volatility or market stress continue to exist (Vidal-Tomás, 2021).

For instance in March 2020 due to the Covid-19 crisis Bitcoin experienced a significant price decline, dropping from \$8,901 on March 8 to \$4,842 by March 12, reflecting rapid shifts in sentiment and market fear and uncertainty. This sharp decline was not isolated to Bitcoin, other financial asset also experienced significant losses during this time. However, Bitcoin's price recovery was rapid in the following months, indicating significant market responsiveness to sentiment and speculative activity (Goodell & Goutte, 2021). Such episodes illustrate Bitcoin's sensitivity to investor sentiment, presenting opportunities for sentiment based quantitative strategies.

These documented inefficiencies directly relate to EMH limitations previously discussed, proving Bitcoin as a relevant context for applying behavioural finance concepts and sentiment analysis in quantitative trading

strategies. Understanding Bitcoin's speculative nature and market structure is thus critical for assessing short-term trading opportunities and evaluating sentiment analysis's practical value in quantitative Bitcoin trading.

2.2 Sentiment Analysis in Financial and Cryptocurrency Markets

Sentiment Analysis is an interdisciplinary field with roots in natural language processing (NLP) and computational linguistics. This involves systematically extracting, quantifying and interpreting subjective information from textual data (Liu, 2012).

With the exponential increase of text-based data, sentiment analysis emerged as a valuable tool, and in the context of financial markets, sentiment analysis examines investor attitudes, emotions, and opinions expressed through various sources, such as social media, news articles, financial forums, and analyst reports, aiming to identify market sentiment and its potential influence on asset prices (Bollen et al., 2011; Tetlock, 2007).

Core methodologies for sentiment analysis in finance include lexicon-based and machine learning approaches. Lexicon-based methods rely on predefined dictionaries or lexicons to quantify sentiment polarity from textual data, while machine learning approaches utilize algorithms to classify sentiment by learning patterns from labelled datasets, potentially capturing complex contextual nuances more effectively (Ravi & Ravi, 2015).

In Cryptocurrency markets, including Bitcoin, sentiment analysis is particularly relevant due to their unique characteristics such as high volatility, speculative investment behaviour, and significant influence of online interactions. (Mai et al., 2018) Given that traditional valuation metrics are often unavailable or less relevant for cryptocurrencies, sentiment becomes a critical indicator for investors and traders in these markets (Phillips & Gorse, 2017).

However, sentiment analysis in cryptocurrency markets faces several challenges. Obstacles include noisy data, bot generated content, manipulations, and the need for real time actionable insights. These complexities highlight the importance of carefully selecting suitable sentiment analysis methods that effectively handle noise and adapt to market conditions. The specific strengths, limitations, and suitability of various sentiment analysis methods are explored in greater detail in section 2.4. It is also essential to understand the diverse data sources from which sentiment, in this case digital, is extracted which will be covered in the next section.

2.3 Data Sources

Reliable sentiment analysis depends heavily on the quality, availability, and suitability of data sources. With the increasing number of participants on social media platforms and increasing search volume, the variety and accessibility of sentiment data significantly increased over the past few years (Xu et al., 2022). The data sources for digital sentiment analysis can be broadly categorized into publicly available and paid sources.

This section of the literature review examines the characteristics, advantages, and limitations of these data sources, providing key insights for the extraction and quantification of sentiment data in the context of quantitative Bitcoin trading strategies.

2.3.1 Publicly Available Sources

Publicly available sources offer cost effective access and wide accessibility to large volumes of sentiment data. However, extracting meaningful signals often requires significant preprocessing and filtering. Key publicly available sentiment sources include social media platforms, public search volumes, and financial news websites.

Social media platforms, particularly X (Twitter) and Reddit, have emerged as major sources of sentiment data for cryptocurrency markets. Kraaijeveld & De Smedt (2020) found that Twitter (X) sentiment has a significant impact on Bitcoin returns and trading volumes, highlighting its relevance for trading strategies. Reddit, with its community focussed model through dedicated subreddits to Cryptocurrencies, provides in depth discussions and sentiment indicators relevant to investor sentiment (Hurjui, 2022).

Nonetheless, accessing these platforms has recently become more challenging due to API restrictions and pricing strategies introduced by Twitter (Mehta, 2023) and Reddit (Wiggers, 2023). Consequently, researchers and traders frequently resort to web scraping, which, while feasible, is resource-intensive and requires substantial data cleaning efforts to ensure quality and reliability.

Financial news websites and specialized cryptocurrency news outlets, such as CoinDesk, serve as valuable sources of sentiment data. These platforms provide current news, in-depth analyses, and expert opinions that can influence investor sentiment and market dynamics.

Furthermore, Google Trends data, reflecting search interest in Bitcoin related terms has been shown to correlate with Bitcoin price movements. Kristoufek (2013) demonstrated that increased search volume for Bitcoin-related terms often come before price increases, suggesting its potential as a sentiment indicator.

While these publicly available sources offer extensive data, they present challenges such as noise, potential manipulation, and the need for sophisticated processing techniques to extract meaningful sentiment signals. On the other hand, their accessibility and real-time nature make them valuable resources for sentiment analysis in Bitcoin trading strategies.

2.3.2 Paid Data Sources

Paid sentiment data sources typically offer structured, pre-processed and professionally analysed sentiment data. Providers, such as EODHD, Santiment and RavenPack specialize in sentiment analytics tailored specifically to financial markets and cryptocurrencies. These vendors aggregate and process vast amounts of textual data from diverse sources, ranging from news websites and social media to proprietary datasets (EODHD, 2025; Santiment, 2025; RavenPack, 2025).

The primary advantages of paid sentiment analytics include higher data quality, reduced noise, and access to specialized sentiment metrics designed explicitly for trading strategies. Professional providers frequently utilize proprietary algorithms and machine-learning techniques to filter irrelevant information, resulting in sentiment data that is immediately actionable. However, paid sources also present significant limitations, most notably their cost, which can be too high for academic research or exploratory analyses. Furthermore, the exact methodologies underlying sentiment scores are often proprietary, reducing transparency and potentially limiting methodological flexibility for exploratory research.

Given these practical constraints, in particular cost, accessibility, and the methodological transparency, this thesis primarily leverages publicly available sources for sentiment data. Awareness of paid sources remains important for comparative purposes, as well as potential future research directions.

2.4 Sentiment Analysis Techniques for Bitcoin Trading

Choosing appropriate sentiment analysis techniques is crucial for extracting meaningful insights from textual data in cryptocurrency markets. Sentiment analysis methods vary widely, from straightforward lexicon based approaches relying on predefined sentiment dictionaries, to more sophisticated natural language processing (NLP) and machine learning models capable of capturing in depth context. The effectiveness of these methods differs significantly regarding accuracy, computational complexity, real-time applicability, and suitability for an exploratory trading prototype. The following subsections provide an overview of the most relevant techniques, discussing their practical advantages and limitations in the context of quantitative Bitcoin trading.

2.4.1 Lexicon Based and NLP Methods

Lexicon based approaches utilize predefined sentiment dictionaries or lexicons, where individual words or phrases are associated with sentiment scores indicating positive, negative, or neutral sentiment (Taboada et al., 2011). In practice, textual data such as social media posts or news articles, is scanned and sentiment scores are computed based on aggregated values of matched words. Commonly used Sentiment lexicons include VADER and SentiWordNet, each designed for specific contexts (Esuli & Sebastiani, 2006; Hutto & Gilbert, 2014).

One particularly suitable lexicon based method for analysing sentiment in social media and online communication is VADER, introduced by C. Hutto and Gilbert (2014). VADER is designed to handle social media text,

effectively capturing expressions of sentiment common on platforms such as Twitter or Reddit, including emojis, slang, and informal language. It generates sentiment scores based on a comprehensive lexicon and considers intensity modifiers, punctuation, capitalization, and negation. VADER's computational efficiency, ease of implementation, and demonstrated effectiveness in social-media-driven contexts make it particularly appropriate for exploratory sentiment analysis within Bitcoin trading strategies. C. Hutto and Gilbert (2014) made VADER open source and accessible on their GitHub page.

Natural Language Processing (NLP), more broadly, includes computational techniques designed to analyse, understand, and interpret human language. Lexicon based sentiment analysis methods, such as VADER, represent a fundamental application within NLP, explicitly focusing on extracting subjective sentiment information from textual data. While NLP methods can vary significantly in complexity, ranging from simple lexicons to advanced machine learning models, they collectively enable the structured interpretation of unstructured text, thereby facilitating quantitative analysis of sentiment signals.

Despite their advantages of simplicity, interpretability, and efficiency, lexicon-based methods often struggle with context-dependent nuances such as sarcasm, slang, or complex linguistic structures commonly encountered in cryptocurrency-related discussions (Hutto & Gilbert, 2014). These limitations can be addressed by more advanced NLP techniques, particularly machine learning approaches, as discussed in the following section.

2.4.2 Machine Learning Based Methods

Machine learning-based sentiment analysis methods have increasingly gained importance due to their capability to handle linguistic nuances and context more effectively than lexicon based approaches. These methods use supervised learning algorithms, including traditional classifiers such as Naive Bayes, Support Vector Machines (SVM), and particularly, more advanced neural network models (Zhang et al., 2018). Among the most significant recent advancements are transformer based models like BERT (Bidirectional Encoder Representations from Transformers), introduced by Devlin et al. (2019). BERT leverages bidirectional contextual embeddings, significantly enhancing the model's understanding of the complex contextual relationships in textual data, resulting in superior sentiment classification performance.

For financial and cryptocurrency specific contexts, specialized BERT variants have been developed, such as FinBERT, fine-tuned explicitly on financial text data. FinBERT incorporates domain-specific vocabulary and linguistic patterns found in financial discourse, thereby offering improved accuracy and interpretability for financial sentiment tasks compared to standard BERT models (Araci, 2019). This specialization enables FinBERT to better interpret market-related discussions, analyst reports, and social media texts related to cryptocurrencies like Bitcoin.

Although their advantages in contextual accuracy, machine learning-based methods like BERT and FinBERT require substantial computational resources, extensive labelled datasets for training. However, their potential in delivering highly accurate and contextually nuanced sentiment signals makes them powerful candidates for quantitative Bitcoin trading strategies.

2.4.3 Comparative Analysis and Practical Suitability

The previous sections outlined two primary categories of sentiment analysis methods relevant to quantitative Bitcoin trading strategies: lexicon based NLP methods, represented by VADER, and advanced machine learning approaches, exemplified by transformer based models such as BERT and its financial domain-specific variant, FinBERT. Each category exhibits distinct advantages and constraints.

Machine learning methods, notably BERT-based models, offer substantial advantages in accurately capturing complex linguistic contexts and domain-specific nuances due to their advanced natural language processing capabilities (Devlin et al., 2019; Araci, 2019). Specifically, FinBERT's specialization for financial texts can further enhance sentiment classification accuracy in cryptocurrency market analyses. However, these methods require considerable computational resources, labelled datasets for effective training and fine tuning, and technical expertise. Such requirements can pose practical constraints in exploratory research contexts or for rapid, resource limited prototyping scenarios.

Lexicon based methods, such as VADER, contrast sharply by providing computational simplicity, rapid implementation, and ease of interpretation (Hutto & Gilbert, 2014). These methods are particularly suitable for exploratory analyses, as they are well suited to processing large datasets quickly and generating easily interpretable sentiment scores. Although they are inherently less sophisticated in capturing complex linguistic contexts compared to transformer based models, their advantages in computational efficiency, ease of use, and real time applicability make them ideal initial candidates for prototype-level sentiment based trading strategy development.

Considering the exploratory scope of this thesis, practical constraints such as limited computational resources, and the need for efficient experimentation, this thesis leverages a combined sentiment extraction approach integrating both lexicon-based (VADER) and machine learning-based methods (FinBERT). Utilizing VADER provides computational efficiency, ease of implementation, and rapid interpretability of sentiment signals, making it suitable for quick exploratory analyses. Simultaneously integrating FinBERT capitalizes on its advanced natural language processing capabilities and specialized accuracy for financial contexts, enhancing the robustness and reliability of sentiment signals. This combined methodological approach ensures comprehensive coverage, capturing diverse sentiment dynamics while addressing practical limitations and delivering richer insights into sentiment driven Bitcoin trading strategies.

2.5 Bitcoin Trading Strategies

A trading strategy can be defined as a structured approach involving clear rules or criteria that guide investment decisions, such as when to buy, sell, or hold a financial asset (Ernest P. Chan, 2021). Commonly known basic strategies include buy-and-hold, simply purchasing an asset and maintaining the position over an extended period, and dollar-cost averaging (DCA), systematically investing equal amounts at regular intervals to reduce price volatility impact (Milevsky & Eposner, 2011). While these fundamental strategies are popular and easy to implement, quantitative trading strategies apply statistical, computational, or data driven methods to identify market opportunities systematically (Ernest P. Chan, 2021).

Bitcoin's unique market characteristics create both challenges and opportunities for trading strategy implementation. Traditional strategies like buy-and-hold have shown effectiveness during Bitcoin's long-term price appreciation periods, while dollar-cost averaging has gained popularity as a method to mitigate the impact of Bitcoin's extreme price swings (Hudson & Urquhart, 2021). However, the market dynamics of cryptocurrencies have also encouraged innovation in more sophisticated approaches already known in traditional assets. Quantitative strategies utilizing technical indicators, ML models leveraging pattern recognition, and sentiment driven approaches analysing social media and news data have all emerged as potential methods to capture alpha in Bitcoin markets.

This section reviews these quantitative and machine learning-based trading strategies, before providing an in-depth exploration of sentiment driven strategies. The main aim is to understand how digital sentiment indicators can practically inform and enhance trading decisions, particularly in generating alpha or improving risk-adjusted returns.

2.5.1 *Quantitative Trading Strategies*

Quantitative trading strategies systematically use statistical, mathematical, and computational techniques to identify profitable market opportunities and make investment decisions (Chan, 2021). These strategies rely primarily on historical price data, market indicators, or fundamental data to generate actionable trading signals, which are subsequently executed either algorithmically or through clearly defined decision rules.

Common quantitative approaches applied to Bitcoin trading include trend-following, mean reversion, and arbitrage strategies (Ahmed et al., 2020; Petukhina et al., 2018). Trend-following strategies assume that existing price trends will persist, utilizing technical indicators such as moving averages, Relative Strength Index (RSI), or Momentum to identify entry and exit points (Corbet et al., 2019). Conversely, mean reversion strategies are based on the expectation that prices eventually revert to historical averages after significant deviations, employing indicators such as, simple or exponential moving averages, Bollinger Bands or statistical thresholds to detect overbought or oversold conditions.

Arbitrage strategies exploit price differences across multiple cryptocurrency exchanges, which can occur frequently due to fragmented liquidity and asynchronous market dynamics (Makarov & Schoar, 2020). Although theoretically risk-free, arbitrage in cryptocurrency markets involves challenges such as transaction costs, market illiquidity, and execution speed limitations.

Despite their systematic nature and popularity, purely quantitative strategies face limitations in Bitcoin markets, particularly when sudden sentiment driven events or emotional market reactions disrupt historically based patterns. Consequently, integrating sentiment indicators or alternative data sources may offer significant opportunities to improve traditional quantitative trading approaches.

2.5.2 *Machine learning based strategies*

Machine learning based strategies have increasingly been applied to cryptocurrency trading, leveraging computational algorithms and statistical techniques to analyse historical price data and identify predictive patterns or signals. Unlike traditional quantitative approaches, ML methods dynamically adapt to market conditions by learning from data, potentially offering enhanced predictive capabilities (Fang et al., 2021).

Common machine learning techniques utilized in Bitcoin trading include supervised learning methods such as Support Vector Machines, Random Forests, and neural network-based models, including advanced architectures like Long Short-Term Memory and Deep Reinforcement Learning (Fischer & Krauss, 2018; Jiang et al., 2017). These methods typically focus on predicting future price movements, volatility patterns, or identifying optimal trading decisions based on historical market data.

Recent research have extended ML strategies further by incorporating alternative data sources, such as investor sentiment from social media, news sentiment, or market microstructure data, significantly enhancing their predictive performance. For example, multi level deep reinforcement learning frameworks have been developed, integrating historical price data with digital sentiment indicators to optimize trading strategies specifically within the Bitcoin market (Otabek & Choi, 2024).

Regardless of promising predictive capabilities, machine learning approaches present significant challenges, including high computational complexity, large data requirements, being prone to overfitting, and interpretability issues. Therefore, cautious implementation and thorough evaluation are essential when deploying ML-based strategies.

Given these considerations, the next section will explicitly examine sentiment driven trading strategies, including both their practical implementation and their potential advantages over purely historical-data-driven ML methods.

2.5.3 Sentiment Driven Trading Strategies

Sentiment driven trading strategies incorporate investor sentiment data derived from textual analysis of digital sources, such as social media platforms, search volume, and online news articles, into quantitative trading models. Unlike traditional quantitative or only price based machine learning methods, these strategies seek to exploit market inefficiencies created by emotional biases, herd behaviour, or speculative sentiment widespread in cryptocurrency markets (Garcia & Schweitzer, 2015; Mai et al., 2018).

the conceptual basis of sentiment driven strategies involves capturing, quantifying, and translating investor emotions into trading signals. Sentiment scores derived through NLP techniques, such as lexicon based or transformer based machine learning methods, provide quantitative measures of market sentiment (Hutto & Gilbert, 2014; Araci, 2019). These scores can be integrated directly into trading models as indicators to trigger buy, sell, or hold decisions.

Practical integration involves defining thresholds for sentiment scores. For instance, positive sentiment above a defined threshold might trigger buy signals, while negative sentiment below a certain threshold could initiate sell or exit signals (Phillips & Gorse, 2017). Additionally, sentiment indicators can be combined with other quantitative indicators (e.g., volatility measures, momentum signals) to enhance signal robustness.

Despite their conceptual possibilities, sentiment driven strategies face challenges. Extracting accurate and actionable sentiment data is complex due to noisy textual inputs, context dependent language, and evolving digital communication styles. Moreover, sentiment indicators may lag actual market moves or reflect manipulative or artificial sentiment signals (e.g., bots, coordinated social media campaigns), potentially reducing reliability (Kraaijeveld & De Smedt, 2020).

Nevertheless, sentiment driven approaches offer potential, particularly in capturing short-term market inefficiencies driven by emotional reactions or speculative trading patterns. Empirical research indicates sentiment indicators can indeed improve trading performance, generate alpha, or enhance risk-adjusted returns when carefully integrated into quantitative models, particularly in highly sentiment sensitive markets such as Bitcoin (Garcia & Schweitzer, 2015; Mai et al., 2018).

3 Methodology

This chapter describes the research methodology used to explore a sentiment driven Bitcoin trading strategy. The chapter outlines the selected methodological approach, defines the research design and details the tools and techniques applied for data collection, sentiment analysis, and strategy evaluation.

3.1 Research approach

The thesis employs a quantitative research approach, aligning with the goal of objectively evaluating the impact of numerical and digital sentiment indicators on Bitcoin trading strategy performance. This approach ensures a systematic and replicable exploration suitable for statistical analysis.

Given the exploratory nature of developing a novel trading strategy prototype, this research adopts an iterative modelling workflow, adapted from common data science and forecasting methodologies (Rob J Hyndman, George Athanasopoulos, 2021). This iterative process, illustrated in Figure 2, provides a structured framework for developing, testing, and refining the sentiment driven strategy.

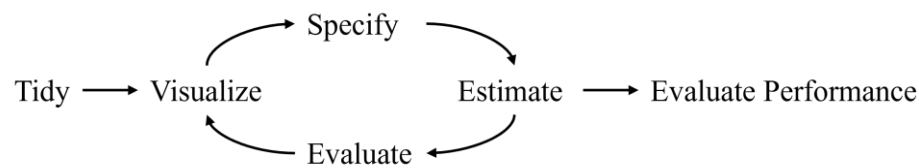


Figure 2: Iterative modelling workflow for exploratory sentiment analysis and quantitative strategy development, inspired by Hyndman & Athanasopoulos (2021)

The power of this workflow lies in its iterative nature. The results from the “Evaluate” stage feed back into the process. Visualising the evaluation results can reveal weaknesses or areas for improvement, leading back to the “Specify” stage to refine the strategy rules or even revisit the “Visualise” stage to explore data relationships further. This loop allows for systematic exploration and refinement, which is essential for developing a prototype in a complex. The final output of this process for the thesis is the comprehensively Evaluated Performance of the developed prototype strategies.

This research prioritises practical applicability over theoretical elegance. Rather than seeking to definitively prove market inefficiency or test rigid hypotheses, the focus is on developing actionable insights for real-world trading applications. This aligns with the exploratory nature of applying sentiment analysis to cryptocurrency markets, where established financial theories may not fully apply and market microstructure differs substantially from traditional assets.

The quantitative approach enables objective evaluation within the defined scope (Chapter 1.3), focusing on measurable performance metrics, Sharpe ratios, maximum drawdowns, and risk-adjusted returns, rather than

subjective assessments of strategy quality. The final output of this process for the thesis is the comprehensively Evaluated Performance of the developed prototype strategy.

3.2 Research Tools and Techniques

This section outlines the methodological framework for data collection, sentiment analysis, and strategy development. The selected tools and techniques are designed to address the research questions while maintaining practical and academic feasibility within the constraints of a bachelor thesis.

3.2.1 Sentiment Analysis Tools

The thesis utilizes a comparative approach using two complementary sentiment analysis methods, described in Table 1, to assess their relative effectiveness for Bitcoin trading applications and use them in symbiosis.

Method	Type	Key Advantages	Limitations
VADER	Lexicon-based	• Social media optimised • Computationally efficient • Transparent methodology • No training data needed	• Limited context understanding • May miss sarcasm/ irony
FinBERT	ML/Transformer	• Domain specialized • Contextual understanding • Superior accuracy	• Computationally intensive • Requires GPU • Less interpretable

Table 1: Sentiment Analysis Tools comparison

Methodological Comparison

These complementary approaches enable comprehensive sentiment analysis: VADER provides rapid, interpretable baseline sentiment scores suitable for real time applications, while FinBERT offers sophisticated contextual analysis for validation and enhanced accuracy. The combination addresses both practical implementation constraints and academic requirements.

3.2.2 Data Collection and Sources

The research methodology requires multiple data sources to capture comprehensive sentiment indicators and enable robust sentiment signals for a trading strategy.

Primary Sentiment Data: Reddit

Reddit serves as the primary sentiment data source based on several methodological considerations. It offers extensive data availability with historical data accessible through academic research channels. Additionally,

Reddit's community focus provides dedicated cryptocurrency subreddits that host concentrated and relevant discussions. The platform also ensures content quality with longer form discussions and less bot contamination compared to other social platforms.

The dataset, which covers the two subreddits r/Bitcoin and r/CryptoCurrency and is used to perform sentiment analysis, was obtained from academictorrents.com. This dataset combines Pushshift API data (pre-April 2023) with community-maintained collection efforts (post April 2023).

Secondary Sentiment Data: Google Trends

Google Trends data provides complementary sentiment indicators, representing potential broader retail investor attention. It offers a broader scope, providing insights beyond the technical or enthusiast communities of Reddit. Additionally, Google Trends ensures real-time accessibility with a publicly accessible API suitable for practical, ongoing implementation. It also has empirical validation, with proven effectiveness in forecasting Bitcoin price movements (Kristoufek, 2013).

The data retrieval process employed the PyTrends unofficial API, implemented with error handling and rate limiting to mitigate API constraints.

Financial News Sentiment

Professional news sentiment from the EODHD Financial API provides an institutional perspective, offering pre-processed sentiment scores from financial news sources. It serves as a comparative baseline, enabling a comparison between social media and professional sentiment. Additionally, it ensures data quality with a structured and reliable data feed suitable for quantitative analysis.

The integration of these multiple sentiment sources is designed to enhance the depth and robustness of the final trading strategy's signals.

Market Data: Bitcoin Prices

Historical Bitcoin price data are essential as the primary dependent variable for strategy development and performance evaluation. Price data were sourced from the EODHD Financial API, chosen for its data quality, which includes institutional grade accuracy with verified daily Open, High, Low, Close, and adjusted closing prices. It also offers comprehensive coverage with a complete daily price history spanning the full analysis period from January 2011 to December 2023. Additionally, the Yahoo Finance API serves as a validation source to ensure the reliability and robustness of the primary data.

This dual-sourced approach ensures robustness and mitigates data risk, supporting reliable backtesting and analysis.

3.2.3 *Data Preprocessing & Feature Engineering*

Both the sentiment feeds and the price series undergo a uniform preprocessing workflow. All records first undergo cleaning and searching of missing values, obvious outliers and duplicated timestamps. Textual sentiment scores will be normalised for an aggregation to remove differences. Subsequently, each sentiment observation is aligned with the corresponding daily Bitcoin close by an exact timestamp join.

Daily Bitcoin log returns are calculated as:

$$r_t = \ln(P_t) - \ln(P_{t-1}),$$

where P_t represents today's closing price, and P_{t-1} the previous day's closing price.

A detailed, step-by-step description of these procedures is provided in Chapter 4.1, where the practical implementation is documented.

3.2.4 *Trading Strategy Development*

Strategy Framework

The prototype trading strategy employs a rule-based approach, integrating sentiment indicators with risk management. Signal generation is based on percentile-based thresholds for sentiment driven buy/sell decisions. Risk management is addressed through position sizing and stop-loss mechanisms. The execution logic follows binary exposure rules, determining whether to be in or out of the market rather than taking leveraged positions.

Implementation Environment

The strategy is implemented using Python for data processing, analysis, and backtesting. The development platform consists of Jupyter Notebooks, which facilitate interactive development and documentation. Key libraries used include Pandas and NumPy for data manipulation, and NLTK and Transformers for sentiment analysis. Version control is managed through GitHub (Appendix C) to ensure code management and reproducibility.

3.2.5 *Strategy Backtesting and Evaluation*

The sentiment based strategy will be backtested against historical Bitcoin price data. The evaluation compares the prototype strategy's performance to benchmark strategies, namely a simple buy-and-hold strategy, dollar-cost averaging, a cryptocurrency index, CCI30 (Rivin & Scevola, 2018), and the S&P 500.

Performance evaluation primarily employs risk-adjusted metrics, including the Sharpe ratio, to measure the return per unit of risk; maximum drawdown, to assess downside risk and loss potential; reflecting the strategy's accuracy or proportion of profitable trades. These metrics will be calculated using custom Python scripts,

leveraging analytical libraries such as Pandas and NumPy, ensuring transparent, consistent, and reproducible evaluation.

Performance evaluation formulas

Performance metrics are computed according to the following standard formulas:

The annualised Sharpe ratio S is calculated as

$$S = \frac{\bar{r}}{\sigma_r} \sqrt{365}$$

Where $\bar{r}$ denotes the mean of the daily log returns and σ_r denotes their standard deviation. A risk-free rate of zero is assumed. Because Bitcoin trades continuously, an annualization factor of $\sqrt{365}$ is used instead of the conventional $\sqrt{252}$ for equities.

The maximum drawdown D_{max} is defined as

$$D_{max} = \max_{t \in [1, T]} \left(1 - \frac{V_t}{\max_{s \leq t} V_s} \right), \quad V_t = \exp \left(\sum_{i=1}^t r_i \right)$$

With V_t representing the cumulative-return series.

4 Prototype development

This chapter presents the implementation of the sentiment driven Bitcoin trading strategy prototype. Building on the methodology outlined in Chapter 3, this part explains the practical execution of data collection, sentiment analysis, and strategy development.

Section 4.1 describes the data processing pipeline that transforms raw social media posts into quantitative sentiment signals. Section 4.2 provides exploratory analysis of the processed data to identify patterns and relationships. Section 4.3 presents the trading strategy prototype.

This exploratory prototype implementation aims to address the research questions while acknowledging the delimitations set. The focus is on demonstrating feasibility and identifying potential value rather than optimizing for maximum performance.

4.1 Data Processing & Feature Engineering

This first part of the prototype development covers the transformation of raw data sources into analysis ready sentiment signals and features for the trading strategy.

4.1.1 Data Collection

Reddit

The Reddit data was obtained from academictorrents.com as a comprehensive dataset covering Bitcoin and cryptocurrency discussions. This public dataset contains the top 40,000 subreddits and upon downloading the data the necessary subreddits can be chosen. The raw data was composed of four initial files covering the submissions and comments of each subreddit (r/CryptoCurrency, r/Bitcoin) from 2021 to 2024.

Metric	r/Bitcoin	r/CryptoCurrency	Combined Total
Total Submissions	387,275	1,259,988	1,647,263
Total Comments	6,521,266	33,154,921	39,676,187
Raw File Size	≈1.85 GB	≈8.15 GB	≈10.81 GB
Missing days	0	0	0

Table 2: Volume of the raw reddit data

Google Trends

To explore other sentiment sources that could potentially enrich the sentiment based trading strategy, this thesis also explores the potential of Google Trends data related to Bitcoin from 2021-2024. A curated list of search terms was selected to represent both positive and negative sentiment indicators.

Sentiment Type	Search terms
Positive	BTC, Bitcoin, buy Bitcoin, crypto bull run, Bitcoin price, should I buy Bitcoin
Negative	sell Bitcoin, crypto crash

Table 3: Search Terms for Bitcoin Sentiment

Data was retrieved using the PyTrends Python library. Since Google Trends restricts full historical data gathering for daily data, the time period was segmented into six month chunks using a custom function. Each keyword was fetched individually at daily resolution by iterating through these time windows, with retry logic and randomized sleep intervals added to mitigate API rate limitations.

The resulting daily interest over time data was concatenated, sorted chronologically, and deduplicated in case of redundant dates between chunks. The final cleaned time series for each keyword was saved in both .csv and .json format for reproducibility and later integration into the sentiment pipeline.

News Sentiment

In addition to Reddit based sentiment and search volume data, this thesis incorporates news sentiment scores sourced from the French financial data provider EODHD. Daily sentiment scores for *BTC-USD* were retrieved via the EODHD REST API for the period from January 2021 to December 2024. The API returned pre processed sentiment values derived from news headlines and articles relevant to Bitcoin, structured as a daily time series.

However, the available data begins in October 2021, which limits the historical coverage relative to other sentiment sources in this thesis.

Price Data

Daily Open, High, Low and Close prices for *BTC-USD* were retrieved via the EODHD REST API for January 1, 2021 to December 31, 2024. The data required no additional preprocessing aside from adjusting the date format to index it later against the sentiment data by the date key.

To confirm data accuracy, the same daily series was fetched from Yahoo Finance using the *yfinance* library. In the next step the comparison was quantified using the correlation coefficient between EODHD and Yahoo Finance closing prices. Given $r=1$ and the absolute percentage of error being 0.0%, EODHD was selected as the primary price source.

4.1.2 Text Data Preprocessing

To ensure accurate sentiment extraction, the Reddit textual data collected from 2021 to 2024 had to go through several preprocessing steps. The pipeline was designed to enhance data quality, relevance and consistency, ultimately being ready for enabling sentiment analysis and signal engineering.

Firstly, the raw textual data was standardised by converting all submissions and comments into lowercase characters. This normalisation ensured consistency and improved the reliability of the sentiment analysis methods employed, which are sensitive to text casing variations.

Subsequently, text cleaning procedures were applied to remove irrelevant content and formatting artifacts, including:

- URLs, hyperlinks, and markdown formatting were stripped from the text to reduce noise.
- Special characters and punctuation marks that do not contribute meaningfully to sentiment scoring were removed.
- Reddit specific entities such as user mentions (e.g., /u/username) and subreddit references (e.g., /r/subreddit) were eliminated to avoid bias or misleading sentiment signals.
- Submissions and comments explicitly flagged as “[removed]” or “[deleted]” by moderators or original authors were excluded due to their absence of meaningful textual content.

A following filtering step specifically targeted the dataset from the r/CryptoCurrency subreddit to enhance relevance. A keyword based filter was implemented to retain only discussions explicitly referencing Bitcoin-related terms such as “bitcoin” and “btc”. This targeted approach significantly reduced the total number of posts as this subreddit also gets a lot of traction from discussion about other cryptocurrencies.

Subreddit	Raw Submissions	Cleaned Submissions	Raw Comments	Cleaned Comments
r/Bitcoin	387,275	384,150	6,521,266	6,336,679
r/CryptoCurrency	1,259,988	167,081	33,154,921	1,872,486

Table 4: Overview of Reddit Data Before and After initial Preprocessing

The cleaned textual dataset (Table 4) provided a robust basis for accurate sentiment scoring using the VADER and FinBERT methodologies.

4.1.3 Sentiment Analysis: VADER

This section covers the implementation steps of the VADER Method, applied to the preprocessed Reddit dataset from Section 4.1.2.

VADER sentiment analysis was chosen due to its robustness and computational efficiency for social media text, as previously discussed in the literature review chapter 2.4.1 Lexicon Based and NLP Methods. The practical implementation used the NLTK library in Python, specifically the *SentimentIntensityAnalyzer()* class from *nltk.sentiment.vader*.

The cleaned textual data, as described in section 4.1.2, served as the direct input for VADER sentiment scoring. For each Reddit submission and comment, four sentiment metrics were generated:

- **Negative:** indicating the degree of negative sentiment present.
- **Neutral:** indicating the neutral sentiment portion.
- **Positive:** indicating the degree of positive sentiment.
- **Compound:** a composite score between -1 (extremely negative) and +1 (extremely positive), summarizing the overall sentiment.

```
from nltk.sentiment.vader import SentimentIntensityAnalyzer
vader = SentimentIntensityAnalyzer()

df['vader_scores'] = df['text_to_analyze'].apply(vader.polarity)
```

Figure 3: Python workflow to generate VADER sentiment scores

The resulting sentiment scores were expanded into individual columns (*vader_neg*, *vader_neu*, *vader_pos*, and *vader_compound*) and merged back into the original DataFrame and saved to a .csv file. The output files were market with a timestamp to ensure organization for the iterative process.

Reddit Score	Date	Text	vader_neg	vader_neu	vader_pos	vader_compound
2	23/03/2023 10:34	what is the reason for this major drop in volume have not seen it this low since months on binance	0.199	0.801	0	-0.5385
4	05/04/2024 12:19	thinking all of us should sell our bitcoin and buy back cheaper	0.095	0.905	0	-0.2732

Table 5: Example of Reddit Data with VADER Sentiment Scores

After the raw VADER scores were merged back into each Reddit DataFrame and written out with timestamped filenames, the next preparation steps were carried out:

Post-processing and Filtering

Rows that had a VADER compound score of 0 (meaning neutral) were removed, since these entries contribute little directional information and tend to “flatten” the daily averages when aggregated. Entries whose text still contained the markers [removed] or [deleted] were also dropped to eliminate bot- or moderator-generated placeholders that VADER can misinterpret as neutral or slightly positive/negative. Columns that were of no value for further processing in the trading strategy were pruned so the output only contains the daily score and the date.

Daily Aggregation

The cleaned DataFrame was then grouped by calendar date and the mean of the remaining compound scores was computed to produce a single “daily_vader_sentiment” value. This step collapsed thousands of individual comments and submissions into one clear, time-aligned sentiment indicator per day.

To maintain interpretability and flexibility, sentiment scores were kept in their original form after daily aggregation. Any further rescaling or normalization required by the analysis or strategy implementation will be explicitly detailed in the respective.

Final Aggregated Output

The final daily series which is trimmed of neutrality, pruned of irrelevant columns and aggregated to a single compound per day, was exported to a .csv. This clean, uniformly scaled sentiment feed becomes one of the core inputs to the prototype trading strategy.

4.1.4 Sentiment Analysis: FinBERT

To complement the lexicon based VADER approach to get sentiment signals, this thesis also applied a machine learning based sentiment analysis approach using FinBERT. This is a specific BERT transformer model that was finetuned for financial text classification. FinBERT was selected due to the domain specific training and strong popularity, on Huggingface the base BERT model reached over 64 million downloads in April 2025, and FinBERT had over 1.5 million downloads. The broader architecture on which BERT models are based is described in Section 2.4.2.

The FinBERT sentiment analysis was implemented using the Hugging Face transformers library, including the AutoTokenizer and AutoModelForSequenceClassification classes. Due to the computational intensity of transformer models, analysis was conducted on Google Colab using a T4 GPU, as execution on local CPU-based environments proved impractical for large-scale inference. The PyTorch framework was used to handle model inference, softmax probability calculations, and tensor management.

```
from transformers import AutoTokenizer, AutoModelForSequenceClassification
import torch
import torch.nn.functional as F
```

Figure 4: FinBERT Python setup

Each cleaned Reddit post was passed individually to the FinBERT model. The output included:

- **finbert_sentiment:** a categorical label (*positive*, *negative*, or *neutral*) based on predicted class
- **finbert_prob_positive:** the model's predicted probability for the *positive* class
- **finbert_prob_negative:** the model's predicted probability for the *negative* class
- **finbert_prob_neutral:** the predicted probability for the *neutral* class

FinBERT produces raw classification scores (logits) for each sentiment class. These scores are passed through the softmax function to compute class probabilities. The sentiment label with the highest probability is then selected as the model's final prediction for that post.

Reddit Score	Date	Text	prob_positive	prob_negative	prob_neutral	finbert_sentiment
2	23/03/2023 10:34	what is the reason for this major drop in volume have not seen it this low since months on binance	0.009215	0.959505	0.031279	negative
4	05/04/2024 12:19	thinking all of us should sell our bitcoin and buy back cheaper	0.051206	0.034745	0.91405	negative

Table 6: Example of Reddit Data with FinBERT Sentiment Scores

Post-Processing and Filtering

After the FinBERT outputs were merged back into each DataFrame, rows with missing or non-meaningful text content (e.g., [removed], [deleted]) were removed. Columns unrelated to further analysis, such as author and link, were dropped for clarity and memory efficiency. The cleaned datasets were saved as timestamped .csv files to ensure reproducibility during iterative development.

Daily Aggregation

To create a daily sentiment time series, a custom sentiment score was calculated for each entry as:

$$score_i = P(positive)_i - P(negative)_i$$

This combined daily sentiment metric enabled further exploratory analyses and strategic assessments.

4.2 Exploratory Data Analysis: Sentiment and Price

Before using the sentiment data from chapter 4.1 directly to create concrete trading rules, it is essential to develop a understanding of the data itself. Exploratory Data Analysis (EDA) therefore serves two purposes. First it acts as diagnostic: by visualising and summarising each series it can be confirmed that the datasets are complete, time-aligned and free of obvious anomalies that could bias the trading strategy. Secondly, it provides interpretability. Simple descriptive statistics, distribution plots and co-movement charts reveal whether and how the price dynamics of Bitcoin appear to react to swings in sentiment patterns, or the absence thereof. Findings observed here guide the choice of lags, threshold percentiles and risk management parameters in the prototype strategy that follows in chapter 4.3.

The goal is not yet to prove causality, but rather to gather the empirical “fingerprints” that will inform the rule design in the next chapter.

4.2.1 Bitcoin Price Trends

Understanding Bitcoin's price behaviour during the analysis period (2021-2024) provides essential context for evaluating sentiment indicators before relating the sentiment to market behaviour.

Price Trend Overview (2021-2024)

The Bitcoin price trajectory (Figure 5) displays distinct market phases characterized by varying volatility and price dynamics:

- **Early-2021 bull-run:** Bitcoin accelerates from roughly US \$30,000 in January to a first peak near US \$64,000 (April 2021), followed by a sharp 50% draw-down during the May “China-mining” sell-off (Shen & Galbraith, 2021).
- **Second ATH and 2022 collapse:** A recovery rally culminates in the all-time-high of $\approx$ US \$69,000 (November 2021). From there, a prolonged bear-leg takes hold, intensified by the Terra/Luna failure (May 2022) and the FTX bankruptcy (November 2022), ultimately bottoming around US \$16,000 (Reuters, 2022).
- **Sideways accumulation:** Throughout the first half of 2023 Bitcoin trades in a low-to-mid-US \$20,000 range with noticeably lower realised volatility.
- **ETF/halving driven repricing:** Price momentum gathers traction after BlackRock’s spot-ETF filing (June 2023) and accelerates into the gray dashed line, which marks the fourth block-reward halving on 20 April 2024. Post-halving, Bitcoin breaks decisively through the previous ATH and ends 2024 just under US \$100,000.



Figure 5: BTC Daily Closing Prices 2021 – 2024

Statistical Analysis of Daily Returns

Table 7 summarises the statistical properties of the daily log-returns, highlighting significant volatility (std. deviation of 3.24%), heavy-tailed returns (kurtosis of 3.49), and a slight negative skew (-0.14).

Min	Median	Mean	Max	Std	Skew	Kurt
-0.174053	-0.000008	0.000793	0.171821	0.032387	-0.139881	3.491121

Table 7: Summary statistics of daily log-returns (2021 - 2024)

Figure 6 visually demonstrates these statistical characteristics, particularly highlighting Bitcoin’s pronounced leptokurtosis and frequent large deviations.

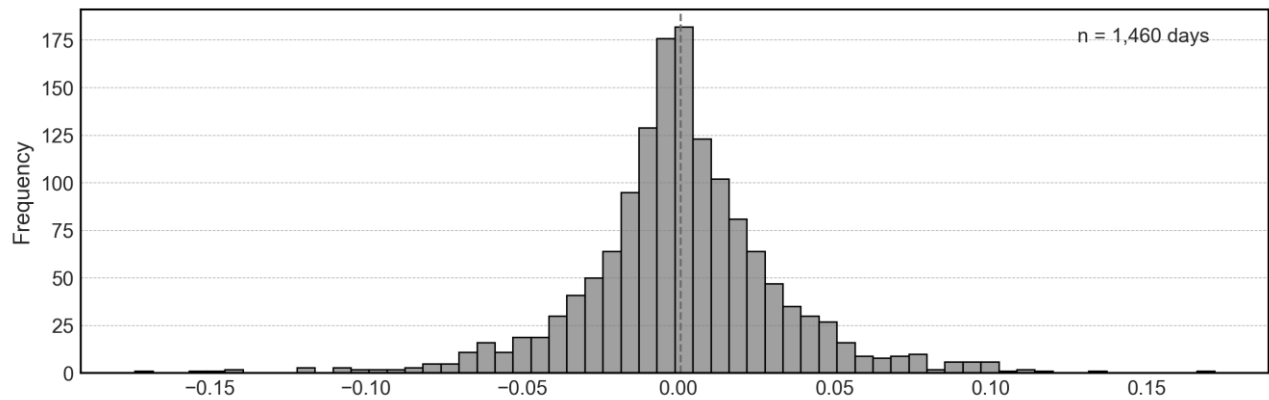


Figure 6: Histogram of daily log-returns (2021-2024)

Rolling Volatility

Given Bitcoin's distinct volatile nature, rolling 30-day volatility analysis was performed to quantify changes across market phases clearly. This approach reveals volatility clusters corresponding to significant macroeconomic and market events (Figure 7).

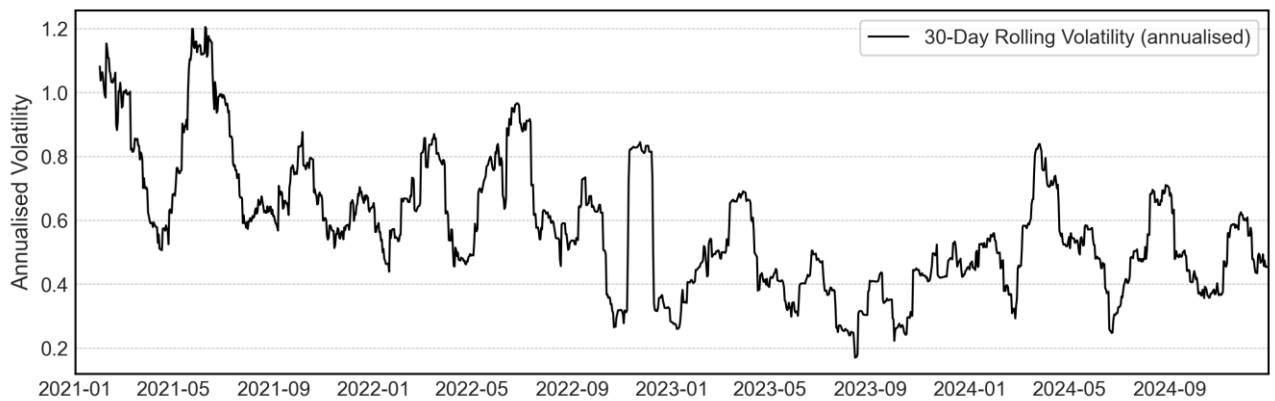


Figure 7: Rolling Volatility 2021-2024

Implications for Strategy Development

Because daily returns are highly leptokurtic (fat-tailed) and occasionally deliver large downside shocks, the prototype in Chapter 4.3 will adopt an un-levered, binary exposure rule (in when sentiment exceeds a dynamic threshold, out otherwise) and a hard stop-loss on open positions. In other words, risk is controlled by time in the market, not by leverage.

4.2.2 Sentiment Series Overview

Following the analysis of Bitcoin's price dynamics, this section investigates sentiment indicators derived from social media data as explanatory variables in the context of prototype trading strategy development.

VADER Sentiment

Figures 8, 9 and Table 8 offer a comprehensive statistical overview of the daily VADER sentiment series, which serves as one of the core trading inputs in the strategy development.

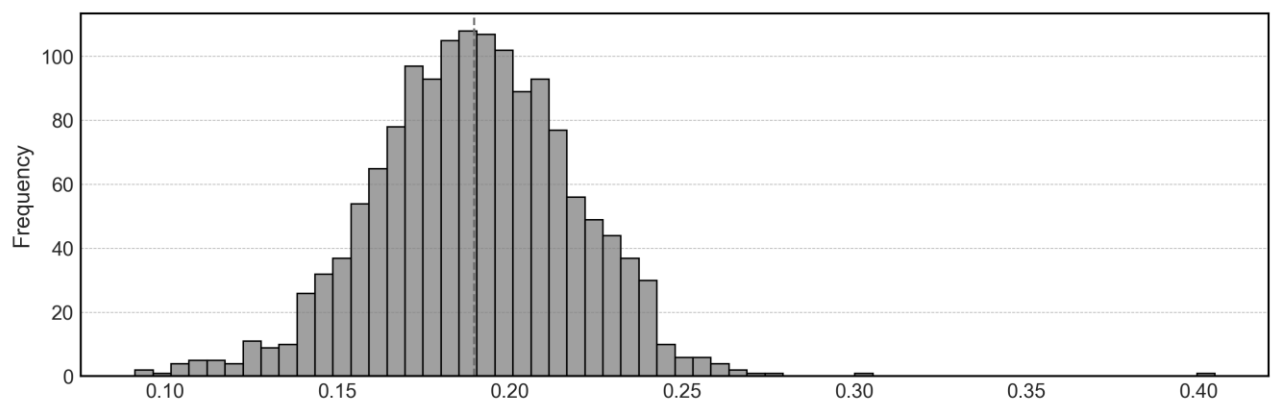


Figure 8: Histogram of daily aggregated VADER sentiment

The distribution in Figure 8 is narrowly centred on a mean and median of ≈ 0.19 , indicating a rather mildly optimistic baseline in Bitcoin related Reddit discussions over 2021-2024.

Table 7 confirms a daily standard deviation of $\sigma \approx 0.029$ excess kurtosis $\kappa \approx 2.0$ (fat-tailed) and a mild positive skew ($\gamma \approx 0.14$).

Min	Median	Mean	Max	Std	Skew	Kurt
0.091232	0.189793	0.189687	0.404624	0.029119	0.139972	2.007667

Table 8: Summary statistics of scaled VADER sentiment

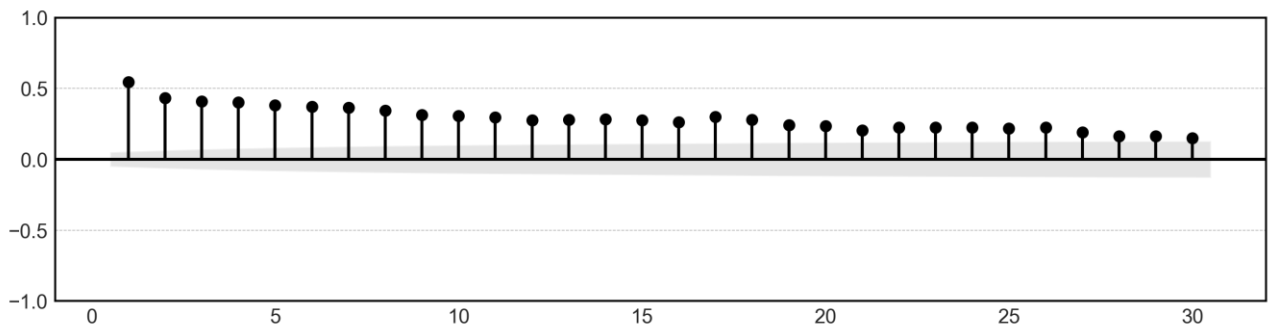


Figure 9: ACF: VADER-scaled Sentiment (lags 1-30)

The autocorrelation function (ACF) in Figure 9 reveals substantial short-run persistence in the VADER sentiment series. Notably, the first-lag autocorrelation exceeds 0.50, gradually declining and approaching insignificance around approximately 25 days. This temporal structure suggests that sentiment shocks typically persist for several consecutive days, thus characterising the series as one with meaningful short-term memory rather than random noise.

Implications for Strategy Development

The presence of a positive bias in the entire series, which sits above zero, means that naïve rules such as “go long when sentiment > 0 ” carry no discriminatory power. Consequently, signals must rely on relative rather than absolute levels.

The existence of fat tails, characterized by periodic sentiment spikes, justifies the use of percentile-based or z-score normalization to isolate extremes while retaining comparability across different regimes. Additionally, the persistence observed in the ACF supports the application of short moving averages or exponentially weighted means. These methods help dampen day-to-day noise without erasing meaningful trends.

These observations inform the signal engineering choices described in Chapter 4.3, where rolling percentiles combined with an exponential smoothing filter are tested as the baseline transformation before back-testing the sentiment driven prototype.

FinBERT Sentiment

The FinBERT model provides a second view of Reddit sentiment. Figures 10 and 11 together with Table 9 describe the daily mean FinBERT compound scores for 2021-2024.

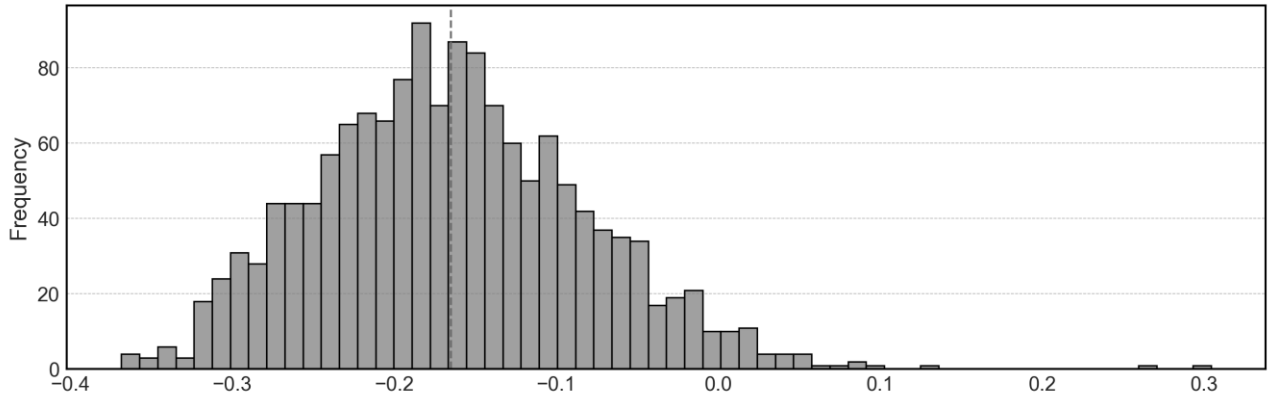


Figure 10: Histogram of daily aggregated FinBERT sentiment

Figure 10 shows a bell-shaped histogram centred left to zero, the mean and median indicate a moderate bearish baseline in the discourse classified by FinBERT. This contrasts with the mildly bullish bias observed in the VADER series.

Skewness $\gamma \approx +0.44$ signals occasional bursts of strong optimism, while excess kurtosis $\kappa \approx 0.77$ points to only mildly fat-tailed behaviour.

Min	Median	Mean	Max	Std	Skew	Kurt
-0.36862	-0.168568	-0.165154	0.304444	0.083256	0.439246	0.767671

Table 9: Summary statistics of scaled FinBERT sentiment

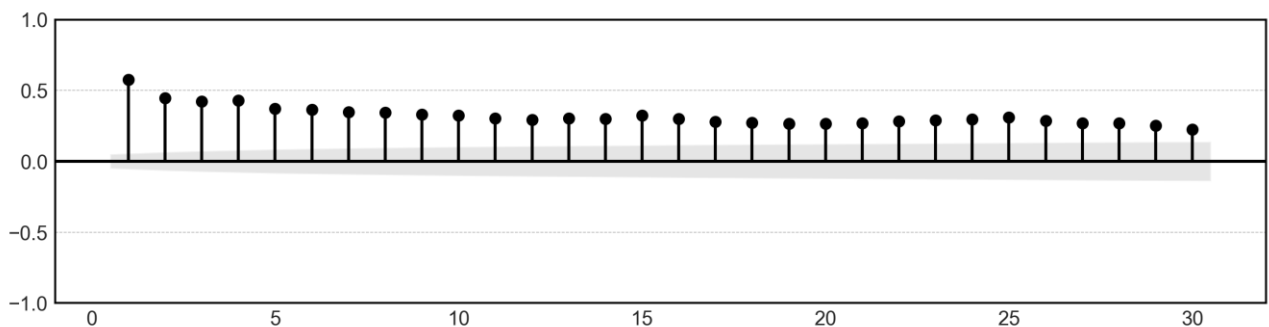


Figure 11: ACF: FinBERT-scaled Sentiment (lags 1-30)

The autocorrelation function in Figure 11 mirrors VADER's memory but at a slightly lower magnitude: the first-lag ACF is about 0.55 and remains significant up to roughly lag 22.

Implications for Strategy Development

The baseline negativity observed in FinBERT scores, which mostly sit below zero, demands that bullish triggers rely on relative positioning, such as being above the 75th percentile, rather than the raw sign of the scores. The persistence in the data supports the continued use of a 3- to 5-day exponential moving average, as long-er windows risk oversmoothing the data.

These findings confirm FinBERT as a complementary sentiment source, contributing a less noisy yet still informative input to the multi-indicator framework adopted in the prototype.

Google Search Volume

Google-Trends data provide a retail attention lens that complements the text-based sentiment series. A list with bullish and bearish related Bitcoin queries was curated and aggregated into a weekly net index to analyse its key properties.

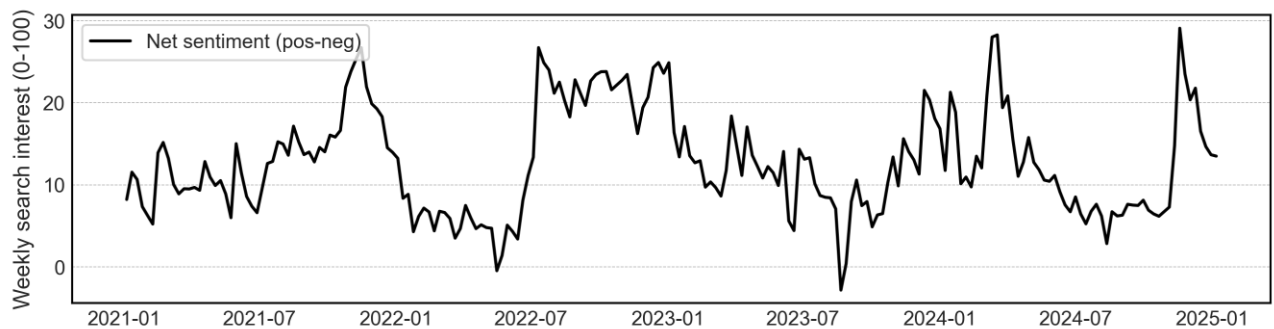


Figure 12: Net Weekly Google Trends Sentiment

Figure 12 plots the weekly net search-interest series (7-day MA). Values fluctuate between 0 and ≈ 30 units. Optimism peaks align with the late-2021 bull run, the BlackRock ETF filing (June 2023), and the 2024 halving, whereas the sharpest dip follows the FTX collapse (November 2022).

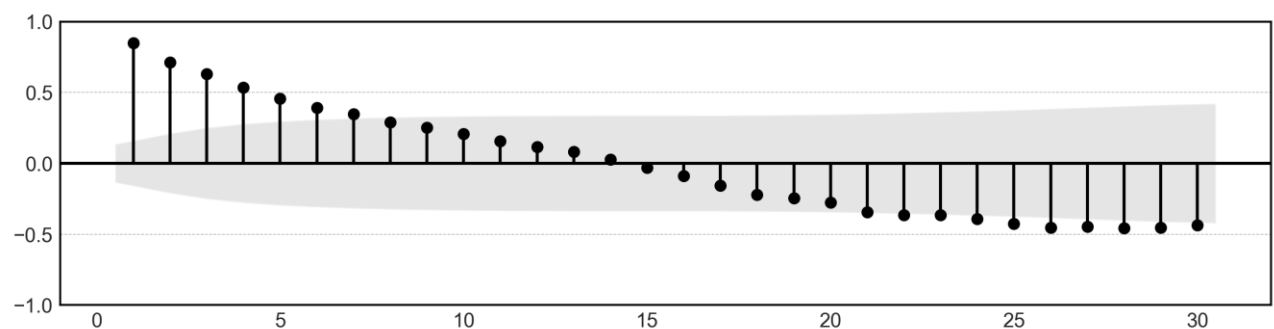


Figure 13: ACF: Weekly Net Google-Trends Sentiment (lags 1–30)

The autocorrelation function in Figure 13 reveals distinct patterns in Google search behaviour. The first-lag autocorrelation is approximately 0.85, substantially higher than the Reddit sentiment series, indicating strong week-to-week persistence in public attention. This persistence gradually decays but remains statistically significant beyond 20 lags, suggesting that shifts in retail interest unfold over extended periods rather than exhibiting the rapid mean-reversion seen in social media sentiment.

Implications for Strategy Development

The Google Trends data show several characteristics that differentiate them from Reddit-based sentiment. There is stronger persistence, with high autocorrelation, suggesting trending behaviour in public attention. Additionally, Google Trends data can act as a lagging indicator, where search volume may reflect rather than predict market movements. Furthermore, it serves as a broader market scale, capturing general public interest beyond just crypto enthusiast communities.

Given these properties, Google Trends data appear better suited as a confirmatory indicator of market regimes rather than as a primary trading signal.

News Sentiment

Professional news sentiment scores from EODHD Financial API were collected to provide an institutional perspective on Bitcoin market sentiment. However, exploratory analysis revealed a critical limitation: the data series begins only in July 2021, missing the first months of the analysis period.

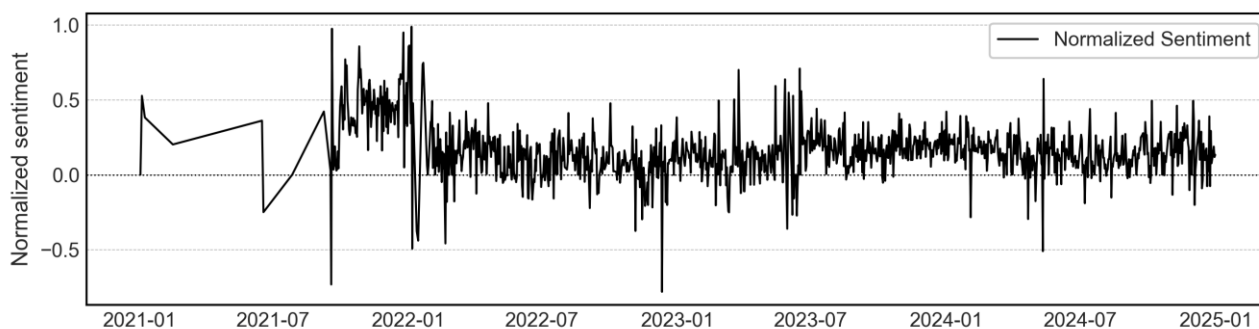


Figure 14: News Sentiment Plot

Figure 14 presents the available news sentiment time series. While the data shows reasonable alignment with major market events, negative spikes during the Terra/Luna and FTX collapses, positive sentiment during recovery periods, the truncated coverage presents methodological challenges:

- **Incomplete IS period:** Missing ~20% of the in-sample training data (Jan-July 2021)
- **Comparison bias:** Performance metrics would not be directly comparable to strategies using full-period data

Given these constraints and the thesis's emphasis on methodological consistency, news sentiment is excluded from the trading strategy development. The complete coverage provided by Reddit sentiment (VADER and FinBERT) ensures robust parameter selection and fair performance evaluation across the entire 2021-2024 period.

4.2.3 Sentiment vs. Price

To empirically establish the potential predictive power of sentiment indicators (VADER and FinBERT) for Bitcoin returns, exploratory analysis on weekly frequency was conducted. Weekly sentiment changes (Δ sentiment) are compared against Bitcoin returns. The purpose is to identify predictive patterns and confirm the practical relevance of sentiment signals, setting a careful foundation for the sentiment driven trading prototype described in Chapter 4.3.

VADER

Figure 15 visually compares weekly changes in VADER sentiment with Bitcoin returns. Although the relationship shows noise, certain peaks and troughs in sentiment head notable price shifts, suggesting a potentially predictive relationship.

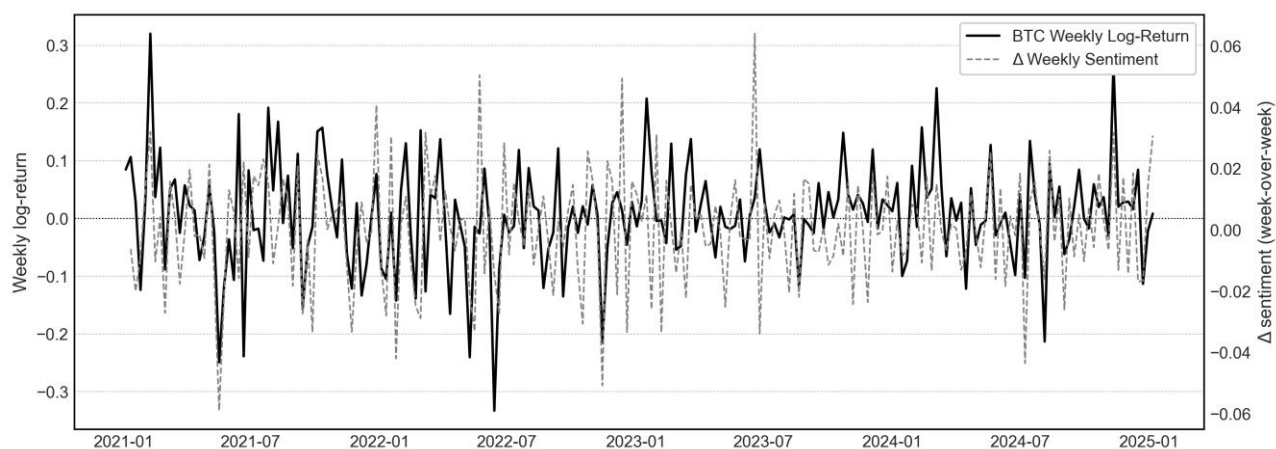
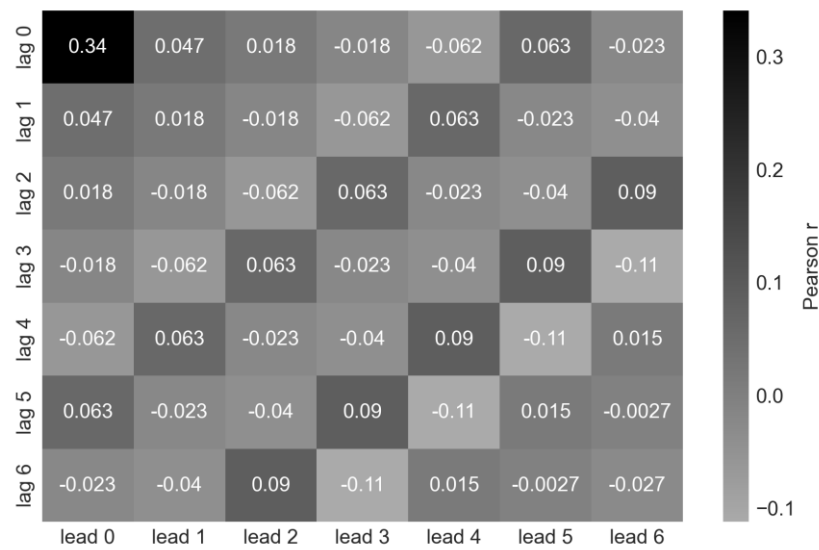


Figure 15: Weekly BTC log-returns against delta VADER Sentiment

The lag-lead correlation heatmap (Figure 16) quantifies these visual patterns and reveals the temporal structure of the sentiment-return relationship. The strongest correlation occurs at lag 0/lead 0 ($\rho = 0.34$), indicating substantial same week co-movement between sentiment changes and Bitcoin returns. This relationship suggests that sentiment shifts and price movements occur largely simultaneously.

Figure 16: VADER Δ sentiment vs BTC returns, lag-lead correlation

While VADER sentiment displays autocorrelation persistence up to 25 days (Figure 16), weekly aggregation was employed in this analysis to reduce visual noise and improve correlation stability. However, meaningful predictive relationships should theoretically only be expected within the first 1-2 weeks, corresponding to sentiment's natural persistence window. Correlations beyond lag 3 weeks likely reflect broader market dynamics rather than sentiment memory effects.

For robustness verification, the same lag-lead analysis was conducted at daily frequency covering lags 0-20 days and leads 0-20 days (Appendix A). The daily correlation map reveals consistent but more granular patterns: the strongest contemporary correlation occurs at lag 0 ($\rho = 0.16$), with modest predictive correlations in the lag 1-3 day range ($\rho \approx 0.05$ -0.07). The daily analysis confirms rapid decay of predictive power, with correlations becoming negligible beyond lag 5-7 days, well within sentiment's persistence window. This daily granularity supports the weekly findings while providing more precise timing for potential trading applications.

Implications for Strategy Development

These findings suggest that VADER sentiment changes are most valuable for identifying contemporary market conditions rather than predicting future returns. The strong lag 0 correlation supports momentum-based trading approaches that utilize current sentiment levels for position sizing and market regime identification. The weak but positive lag 1 correlation provides limited justification for forward-looking strategies, though the effect size may be insufficient to overcome transaction costs in practical implementation.

The rapid decay of predictive power beyond one week reinforces the need for frequent strategy rebalancing and suggests that sentiment based signals should focus on short-term tactical allocation rather than longer-term strategic positioning.

FinBERT

Similarly, Figure 17 plots weekly FinBERT sentiment changes against Bitcoin returns. The visual alignment remains noisy, although occasional synchronization between sentiment and returns is visible.

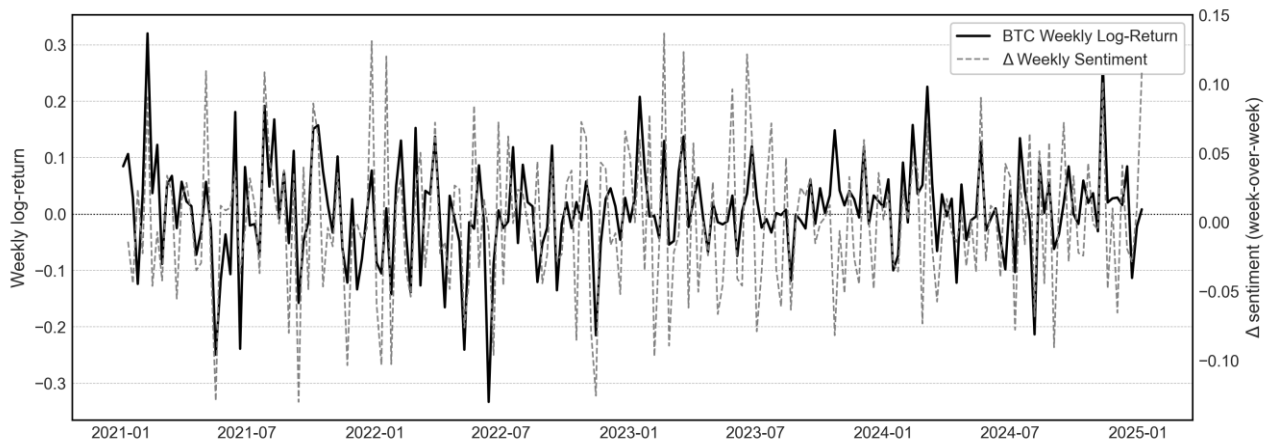


Figure 17: Weekly BTC log-returns against delta FinBERT Sentiment

The lag-lead correlation heatmap (Figure 18) reveals FinBERT's superior contemporary correlation strength compared to VADER. The strongest relationship occurs at lag 0/lead 0 ($\rho = 0.50$), indicating substantial same week co-movement between FinBERT sentiment changes and Bitcoin returns. This correlation is notably stronger than VADER's 0.34, suggesting FinBERT's domain-specific financial training enhances its ability to capture market relevant sentiment shifts.

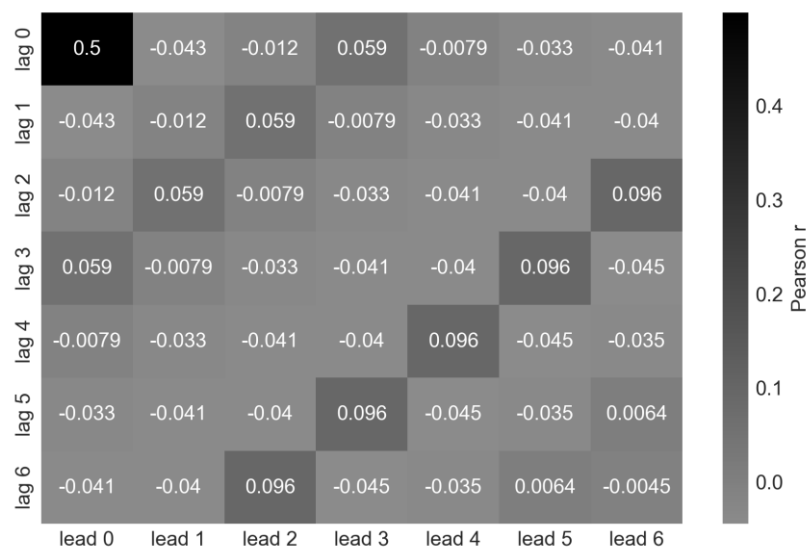


Figure 18: FinBERT Δ sentiment vs BTC returns, lag-lead correlation

However, FinBERT's predictive power beyond the contemporary period is limited. The lag 1/lead 0 correlation is weak and negative ($\rho = -0.043$), contrasting with VADER's modest positive predictive correlation. Modest

positive correlations develop at longer horizons, particularly at lag 3/lead 2 ($\rho = 0.059$) and lag 6/lead 2 ($\rho = 0.096$), though these likely reflect statistical noise rather than meaningful predictive relationships given their sporadic pattern.

The same lag-lead analysis was conducted at daily frequency covering lags 0-20 days and leads 0-20 days (Appendix A). The daily correlation map confirms the primary findings: strong contemporary correlation at lag 0 ($\rho = 0.30$) with rapid decay of any predictive relationships. The daily analysis shows correlations swinging around zero beyond lag 2-3 days, confirming that FinBERT's value lies primarily in contemporary market analysis rather than forward prediction.

Comparison with VADER

FinBERT demonstrates several advantages over VADER in capturing Bitcoin market sentiment:

- **Stronger contemporary correlation:** 0.50 vs 0.34, indicating superior real-time market sensitivity
- **Domain specificity:** Financial training appears to enhance relevance for cryptocurrency discussions
- **Different temporal pattern:** Less persistent predictive correlation but stronger immediate response

However, FinBERT shows weaker forward-looking predictive power, with negative correlation at lag 1 suggesting potential mean-reversion effects that differ from VADER's momentum-like pattern.

Implications for Strategy Development

FinBERT's stronger contemporary correlation, contrasted with its weaker predictive power, suggests that its optimal use differs from that of VADER. FinBERT is particularly valuable for real-time regime identification due to its superior contemporary correlation, making it effective for identifying current market conditions. This characteristic also supports momentum strategies, as the strong lag 0 correlation allows for an immediate response to sentiment shifts. Additionally, FinBERT's better real-time sensitivity can enhance risk management by improving portfolio protection during rapid sentiment deterioration. The different temporal characteristics of FinBERT also suggest potential value in combining it with VADER to achieve comprehensive sentiment coverage.

The absence of meaningful predictive correlation reinforces the need for contemporary rather than anticipatory trading strategies when using FinBERT sentiment signals.

Google Sentiment vs Price

Figure 19 compares weekly changes in Google-Trends net sentiment with Bitcoin returns. As with the Reddit metrics, the series is noisy; nonetheless, spikes in search interest often arise shortly before outsized price moves, hinting at a short horizon predictive connection.

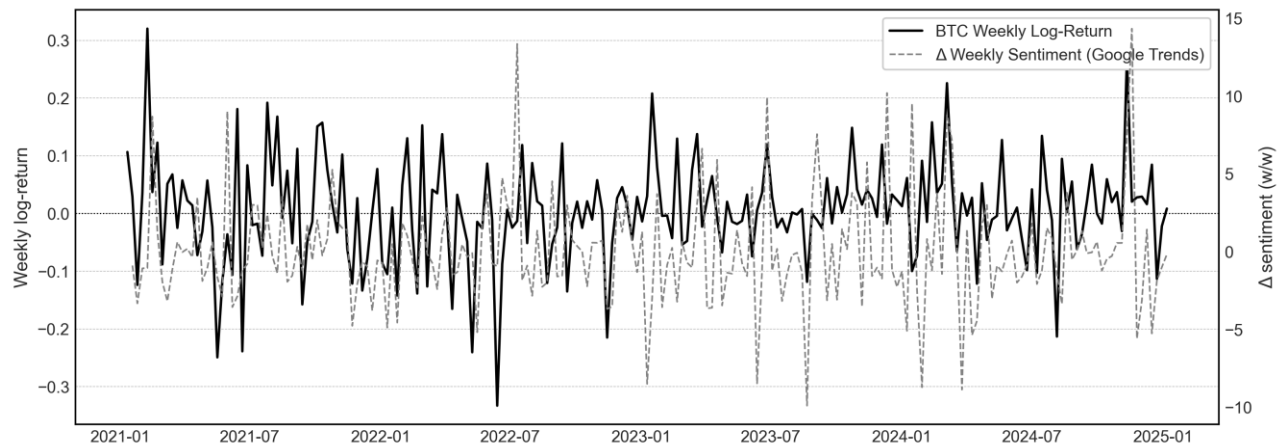


Figure 19: Weekly BTC log-returns against delta Google Sentiment

The lag-lead correlation heatmap (Figure 20) reveals Google Trends temporal relationship with Bitcoin returns. The strongest contemporary correlation occurs at lag 0/lead 0 ($\rho = 0.12$), substantially weaker than both VADER (0.34) and FinBERT (0.50), reflecting Google Trends role as a broader retail attention measure rather than a focused sentiment indicator.

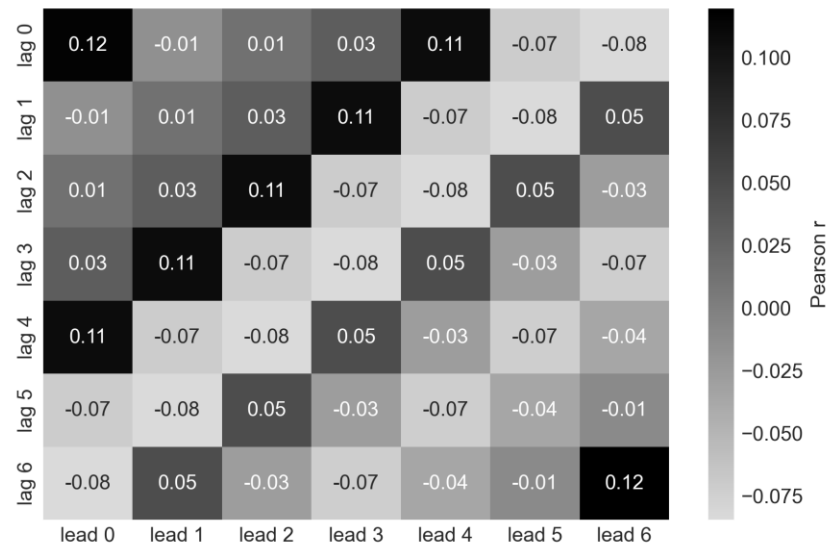


Figure 20: Google Trends Δ-sentiment vs BTC returns, Lag–Lead Correlation Map

Google Trends shows an unusual pattern with correlations of 0.12 at both lag 0 and lag 6, creating a distinctive “diagonal pattern” in the correlation matrix. This suggests cyclical attention patterns rather than actionable trading signals. Forward looking correlations remain weak (0.03-0.05) and inconsistent across lags, indicating limited predictive value.

Implications for Strategy Development

The empirical analysis reveals several factors that disqualify Google Trends as a primary trading signal. The weak signal strength, characterized by a contemporary correlation of 0.12, is insufficient for reliable regime identification. Additionally, Google Trends acts as a lagging indicator, where search volume appears to follow rather than lead price movements. Furthermore, the information provided by Google Trends may be redundant, as the stronger Reddit composite, with a correlation of $\rho = 0.457$, likely captures relevant retail sentiment more effectively.

Given these limitations, Google Trends data will not be incorporated into the trading strategy. The Reddit composite sentiment, with its superior correlation and more direct connection to active market participants, provides a more robust foundation for the sentiment driven approach developed in Chapter 4.3.

Combined Reddit Sentiment vs Price

Figure 21 plots the equally weighted, z-standardized average of VADER and FinBERT sentiment changes against Bitcoin price returns. This Reddit composite sentiment measure combines the lexicon based VADER approach with the ML based FinBERT method, capturing matching aspects of community sentiment. Visually, the composite series produces a smoother signal than the individual components, while preserving the major sentiment swings that align with market movements.

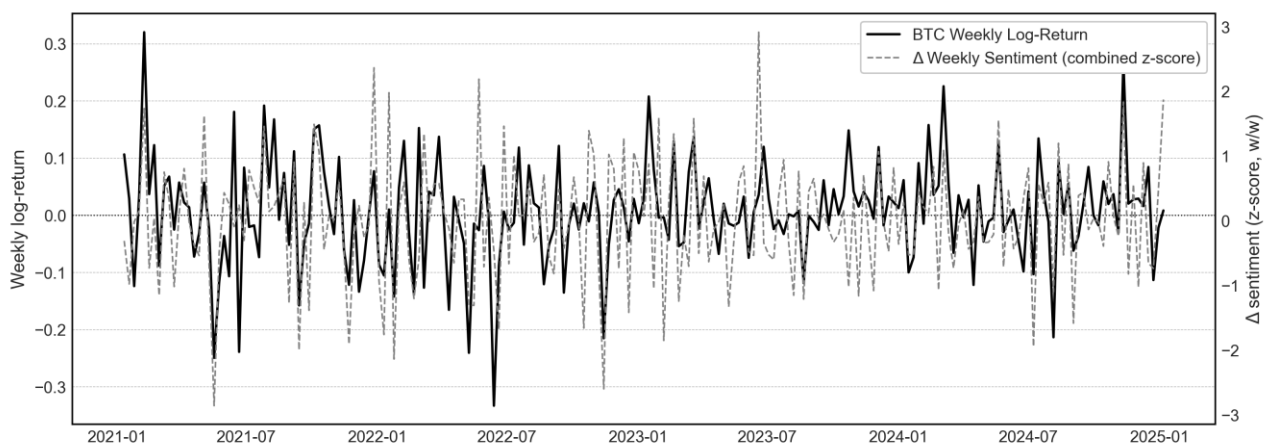


Figure 21: Weekly BTC log-returns against Δ Combined Sentiment (z-scaled average)

The combined sentiment indicator was constructed in a simple equally weight way by:

- Computing weekly sentiment changes (Δ sentiment) for both VADER and FinBERT
- Z-standardising each series to ensure equal weighting regardless of scale differences
- Taking the simple average: $\text{Combined} = (Z_VADER + Z_FinBERT) / 2$

This approach preserves the signal characteristics of both methods, while reducing individual noise present in the individual measures.

The lag-lead correlation heatmap (Figure 22) reveals that the Reddit composite sentiment achieves the strongest contemporary correlation of any individual or combined measure analysed. The peak correlation occurs at lag 0/lead 0 ($\rho = 0.457$).

- Combined Reddit: $\rho = 0.457$ (lag 0)
- FinBERT alone: $\rho = 0.50$ (lag 0)
- VADER alone: $\rho = 0.34$ (lag 0)

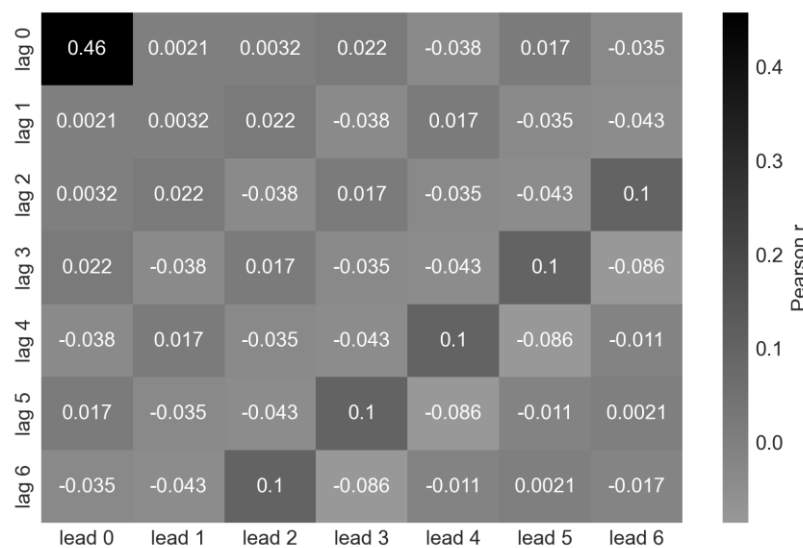


Figure 22: Combined Δ -sentiment vs BTC returns, Lag-Lead Correlation Map

While FinBERT shows slightly higher individual correlation, the composite approach is selected for the trading strategy due to several critical advantages:

- **Risk diversification:** Single-method dependency creates model risk if that method fails
- **Operational robustness:** VADER's computational efficiency ensures real-time viability even if FinBERT processing fails
- **Signal validation:** Agreement between methods increases confidence; divergence flags uncertain conditions
- **Reduced overfitting:** Combining methods reduces the risk of capitalizing on method-specific quirks

The forward-looking correlations remain limited, with lag 1 showing $\rho \approx 0.002$, reinforcing the contemporary nature of sentiment-price relationships. This pattern supports the regime-based trading approach described in Chapter 4.3.

Implications for Strategy Development

The superior contemporary correlation of the composite sentiment, measured at 0.457, strongly justifies its use as the primary trading signal. By combining VADER's computational efficiency with FinBERT's contextual sophistication, a balanced indicator is created. This indicator effectively captures market regime changes, reduces false signals through methodological diversity, and maintains practical implementability. These findings support its use as the primary sentiment filter in Chapter 4.3.

4.3 Strategy

This section translates the empirical insights of subchapter 4.2 into an implementable, risk-managed Bitcoin trading strategy that relies on contemporary sentiment strength rather than forecasting. The prototype remains intentionally simple so that its performance can be credited to the information content of Reddit sentiment.

4.3.1 Strategy Design Framework

Table 10 summarises the three empirical findings from Chapter 4.2.3 and the design implications drawn from each. In essence, Reddit sentiment is correlated with same day returns and forward looking power is limited, thus motivating a risk managed, regime-based allocation rule.

Empirical finding (§ 4.2)	Design implication
Strong same-day correlation between composite Reddit sentiment and BTC return ($\rho = 0.457$).	Exploit <i>current</i> market conditions, not distant prediction. Re-assess exposure at least weekly.
Weak forward correlations (max $\rho = 0.101$ at +3 weeks).	Avoid aggressive directional bets; focus on <i>risk control</i> and rapid de-risking when sentiment deteriorates.
Heterogeneous information sources (Reddit $\approx$ retail, news $\approx$ institutional).	Blend VADER & FinBERT to capture the retail layer; institutional layer remains for future work.

Table 10: Findings of Chapter 4.2

From these observations three guiding principles are derived:

- **Focus on risk management**
Capital preservation outranks absolute return. The strategy must outperform buy-and-hold on Sharpe and max drawdown, even if total return is similar.
- **Regime-based tactical allocation**
Sentiment is used only to classify the market as bullish, neutral or bearish; the portfolio is re-weighted accordingly. No attempt is made to predict the exact magnitude future returns.

- **Conservative implementation**

Position limits lie between 0% and 100% BTC; without leverage and without taking on short positions.

The neutral regime retains modest optimised long exposure to participate in Bitcoin’s upward drift.

4.3.2 *Signal-engineering methodology*

The sentiment driven trading signal is constructed using a structured approach as detailed in Table 11:

Step	Method	Parameter	Justification
(i) Scale alignment	90-day rolling z -score of VADER and FinBERT series.	window = 90	Removes mean/variance differences with no look-ahead.
(ii) Composite signal	Equal-weight average: $\text{sent_raw} = (\text{vader_z} + \text{finbert_z}) / 2$	equal weights	Combines lexicon-based and ML approaches for robustness.
(iii) Smoothing	Exponential moving average (EMA).	span p ; optimised 3 – 10 on IS	Reduces one-day noise while preserving timeliness.
(iv) Regime cut-offs	Rolling 60-day 75 / 25 percentiles of the smoothed series.	window = 60	Data driven thresholds adapt to slow shifts in sentiment distribution.
(v) Position map	Bull = 100 % BTC, Neutral = 10 % BTC, Bear = 0 % BTC (cash).	neutral = optimised 0.1 – 0.40 on IS	Balances upside participation with downside protection.
(vi) Rebalance schedule	First close each week (Monday).	–	Cuts turnover; compatible with retail fee structures (0.10 % assumed).

Table 11: Signal Engineering

The composite sentiment calculation specifically addresses the scale differences between VADER and FinBERT through z -score normalization. The entire signal uses only information available up to time t (2021.01.01 – 2023.12.31), preventing look-ahead bias.

A direct consequence of using these rolling windows is that the strategy requires an initial “warm-up” period (90 days for the longest window) at the start of the backtest. During this period, no trading signals are generated as the indicators gather sufficient data to initialize. This ensures that every signal from that point forward is based on a complete look-back period.

4.3.3 *Parameter Selection & Validation*

To prevent in-sample overfitting, hyperparameters are fixed ex ante based on explicit train-test split:

- **In-sample calibration (IS):** 1 Jan 2021 – 31 Dec 2023 (1096 days)
- **Out-of-sample evaluation (OOS):** 1 Jan 2024 – 31 Dec 2024 (365 days)

Hyperparameter	Candidate set	Selection rule
EMA span pp	{3, 4, ..., 10}	Maximized IS Sharpe ratio (optimal = 5)
Neutral weight	{10%, 20%, 30%, 40%}	Highest IS Sharpe ratio (optimal neutral = 10%)
Percentile thresholds	Fixed at 75th/25th	Standard quartile approach; no optimisation
Volatility target	Median IS volatility ($\approx 47\%$)	Chosen a priori; tested alternatives in robustness analysis

Table 12: Hyperparameters

This structured calibration ensures all parameters are determined ex-ante, preserving the integrity of the OOS evaluation. The choice of a 90-day z-score window reflects the finding that sentiment autocorrelation persists for approximately 25 days (Section 4.2.2), allowing for roughly three cycles within the normalization window.

4.3.4 Strategy Variants

To test robustness and explore risk management enhancements, the thesis evaluates two strategy variants:

1. Strategy A (Sentiment driven):

The primary strategy variant leveraging regime-based tactical allocation, tested separately using:

- Composite sentiment signal (VADER & FinBERT)
- VADER-only sentiment signal
- FinBERT-only sentiment signal

2. Strategy B (Volatility-aware):

An enhanced variant based on Strategy A (composite sentiment), additionally incorporating volatility-based position sizing. Weekly allocations are adjusted according to the volatility scaling factor:

$$scale_t = \min\left(1, \frac{\sigma_{target}}{\sigma_{30,t}}\right)$$

where $\sigma_{30,t}$ represents the rolling 30-day annualised volatility of Bitcoin returns, and σ_{target} is defined as 80% of the median volatility observed in the in-sample period. This mechanism dynamically reduces exposure during periods of heightened volatility, while avoiding leverage (scaling factor capped at 1).

Rationale for Volatility Scaling:

The volatility-aware enhancement addresses a key limitation of pure sentiment based allocation. During high volatility periods, even correct directional calls can result in substantial drawdowns. By scaling position size inversely to volatility, Strategy B aims to maintain consistent risk exposure across different market regimes rather than fixed capital exposure. This approach also seeks to reduce position size during volatile periods

when sentiment signals may be less reliable and gap risk increases. Additionally, it aims to preserve capital during market stress while maintaining upside participation when volatility normalizes.

The 80% provides a conservative buffer, acknowledging that sentiment based strategies may require additional risk reduction compared to passive approaches. This choice reflects the exploratory nature of the research, prioritizing capital preservation while testing the viability of sentiment driven allocation.

Implementation:

This mechanism dynamically reduces exposure during periods of heightened volatility, while avoiding leverage (scaling factor capped at 1). For example:

- When current volatility = 60% and target = 47%, scale = 0.78 (reducing position by 22%)
- When current volatility = 30% and target = 47%, scale = 1.0 (no reduction)

These variants share identical signal-engineering pipelines and parameters fixed in 4.3.3, with no additional OOS parameter optimization performed, ensuring that results remain free of look-ahead bias.

4.3.5 Implementation Considerations

Several practical aspects were considered:

Transaction Costs: The strategy assumes 0.10% round trip transaction costs, consistent with retail tier fees on major cryptocurrency exchanges. Weekly rebalancing was specifically chosen to balance signal responsiveness with cost efficiency, more frequent rebalancing would eat away returns through excessive trading costs.

No Short Selling: The strategy is designed to take only long or cash positions, reflecting practical constraints in cryptocurrency markets where short selling involves several challenges. Engaging in short selling requires margin accounts with additional capital requirements and suffers funding costs. Additionally, there are regulatory restrictions in many jurisdictions that further complicate short selling. The operational complexity associated with short selling also presents additional hurdles.

Risk Management: While no explicit stop-losses are implemented, the regime-based approach provides implicit risk control by moving to cash (0% allocation) when sentiment deteriorates below the 25th percentile. This approach avoids whipsaw losses from tight stops while still providing downside protection.

Market Impact: Given the strategy's modest size assumption (retail-level capital) and weekly rebalancing schedule, market impact is considered negligible. However, scaling to institutional size would require additional considerations including execution algorithms and liquidity analysis.

These design choices prioritize risk-adjusted returns over absolute performance. The conservative approach, unlevered positions, cash allocation during uncertainty, volatility scaling, may underperform during strong

trending markets but should demonstrate superior risk-adjusted metrics across full market cycles. Chapter 5 evaluates whether these theoretical advantages translate into practical outperformance.

5 Backtesting & Results

This chapter evaluates the performance of the sentiment driven trading strategies specified in Chapter 4 through systematic backtesting across both in-sample (IS) and out-of-sample (OOS) periods. By maintaining strict separation between parameter selection (IS: 2021-2023) and performance evaluation (OOS: 2024), the analysis ensures methodological integrity and guards against overfitting bias. The chapter addresses three key objectives:

- **Assess sentiment based alpha generation:** Determine whether regime classification using Reddit sentiment can generate superior risk-adjusted returns compared to passive benchmarks (buy-and-hold, dollar-cost averaging).
- **Validate the composite approach:** Compare the performance of the combined VADER-FinBERT signal against individual sentiment engines to verify whether methodological diversity enhances strategy robustness.
- **Evaluate risk management enhancements:** Quantify the added value of volatility-aware position sizing (Strategy B) in reducing drawdowns while maintaining return potential.

Detailed analyses, including cumulative returns, Sharpe ratios, volatility and maximum drawdown and will directly address the research questions established at the beginning of the thesis, providing transparent and actionable insights into the practical utility and limitations of sentiment driven quantitative Bitcoin trading strategies.

5.1 Backtesting Methodology

5.1.1 Temporal Structure and Data Splits

The backtesting framework employs a walk-forward analysis approach with clear temporal boundaries:

- **In-Sample Period:** 1 January 2021 – 31 December 2023 (1,096 trading days)
 - Used exclusively for parameter selection (EMA span, neutral weight)
 - Encompasses full market cycle: bull market (2021), bear market (2022), recovery (2023)
- **Out-of-Sample Period:** 1 January 2024 – 31 December 2024 (365 trading days)
 - Pure performance evaluation with frozen parameters
 - Tests strategy robustness during new market regime (post-halving bull market)

While IS results are reported for completeness and to verify parameter selection logic, all performance claims and conclusions are based solely on OOS results. IS metrics serve only to demonstrate that the optimization process functioned correctly and did not produce unreliable solutions.

5.1.2 Execution Assumptions

The backtesting engine simulates realistic trading conditions:

- **Price Data:** Daily closing prices for BTC-USD from EODHD
- **Trade Execution:** Weekly rebalancing at Monday close prices
- **Transaction Costs:** 0.10%, representative of retail tier fees on major exchanges
- **Slippage:** Not modelled given Bitcoin's deep liquidity
- **Initial Capital:** Normalized to 1.0 for return calculations

5.1.3 Benchmark Strategies

Three benchmark strategies provide context for performance evaluation:

- **Buy-and-Hold (B&H):** Full investment in Bitcoin from day 1, no rebalancing.
- **Dollar-Cost Averaging (DCA):** Weekly purchases of equal amounts, matching strategy rebalance frequency.
- **S&P 500:** Investment in the S&P 500 index, representing a traditional equity market benchmark with no rebalancing.
- **CCi30:** Market capitalisation weighted index of the 30 largest cryptocurrencies, representing broad-market digital-asset exposure. Held passively to mirror the index's own monthly adjustment schedule.

5.1.4 Performance Metrics

The evaluation employs standard quantitative finance metrics:

Return Metrics:

- **Total return:** Final portfolio value divided by initial value, minus one
- **Annualized volatility:** Daily standard deviation multiplied by $\sqrt{365}$

Risk Metrics:

- **Maximum drawdown:** Largest peak-to-trough decline in cumulative equity
- **Sharpe ratio:** Annualized return divided by annualized volatility (assuming risk-free rate = 0)

These metrics enable direct comparison between the sentiment driven strategies and buy-and-hold benchmark, with particular emphasis on risk-adjusted performance through the Sharpe ratio.

5.1.5 Parameter Selection Process

The strategy optimization involves two hyperparameters, both selected through grid search maximizing in-sample Sharpe ratio:

1. **Neutral Allocation Weight:**

- Grid search over {0.1, 0.2, 0.3, 0.4}
- For each neutral weight, the optimal EMA span is determined
- Optimal value: 10% (neutral_w = 0.1), achieving IS Sharpe of 0.22

2. **EMA Span:**

- Grid search over {3, 4, ..., 10} for each neutral weight
- Optimal value: 5 periods (with neutral_w = 0.1)

Neutral Weight	Best EMA	IS Sharpe	OOS Sharpe
0.1	5	0.22	2.33
0.2	5	0.20	2.29
0.3	5	0.18	2.23
0.4	8	0.17	2.16

Table 13: Neutral Allocation & EMA, Grid Search Results

The grid search reveals that lower neutral allocations (10-20%) consistently outperform higher allocations in both IS and OOS periods. The optimal combination of neutral_w = 0.1 and EMA span = 5 maximizes IS Sharpe at 0.22.

Fixed Parameters (not optimized):

- Percentile thresholds: 75th/25th
- Rolling windows: 90 days (z-score normalization), 60 days (percentile calculation)
- Transaction costs: 0.10%

This minimal optimization approach reduces overfitting risk while allowing the strategy to adapt to Bitcoin's evolving volatility regime.

5.2 Empirical Results

This section systematically presents the empirical performance of the sentiment driven prototype strategies, focusing on their effectiveness in generating alpha, reducing volatility, and managing downside risk. Results are evaluated separately for the in-sample and out-of-sample periods.

5.2.1 In-Sample Performance (2021-2023)

Table 14 summarizes the in-sample performance, which contains diverse market conditions including the 2021 bull market, the 2022 bear market, and the 2023 recovery phase.

Sample		Total Ret	Ann Vol	Sharpe	Max DD
IS	Strategy A (comp)	0.21	0.28	0.22	0.59
	Strategy B (comp)	0.08	0.24	0.1	0.57
	Bitcoin	0.46	0.65	0.19	0.77

Table 14: Performance Summary (IS)

The composite Strategy A achieves moderate returns (121%) compared to Bitcoin buy-and-hold (146%) but notably reduces annualized volatility (from 65% to 28%) and maximum drawdown (from 77% to 59%). Strategy B further lowers volatility (25%) and drawdown (57%) at the expense of reduced total returns (108%), indicating its effective defensive characteristics.

The maximum drawdown reduction is also remarkable: while Bitcoin experienced a 77% decline during the 2022 bear market, Strategy A limited losses to 59% and Strategy B to 57%, demonstrating effective downside protection.

5.2.2 Out-of-Sample Performance (2024)

Table 15 and Figure 23 present the crucial out-of-sample results, representing the true test of strategy viability with frozen parameters in an unseen market environment.

Sample		Total Ret	Ann Vol	Sharpe	Max DD
OOS	Strategy A (comp)	1.43	0.33	2.7	0.14
	Strategy B (comp)	1.41	0.31	2.83	0.12
	Bitcoin	1.21	0.53	1.49	0.26

Table 15: Performance Summary (OOS)

Both strategies significantly outperform the buy-and-hold benchmark during the out-of-sample period, achieving higher total returns (Strategy A: 143%, Strategy B: 141% vs. 121%) and dramatically improving risk metrics. Sharpe ratios for Strategy A (2.70) and Strategy B (2.83) significantly exceed the benchmark (1.49), while maximum drawdowns are halved (14% and 12% vs. 26%).

This performance is particularly notable given the market conditions of 2024, characterized by strong bullish sentiment post-Bitcoin halving, an environment typically favourable for passive buy-and-hold strategies. The enhanced risk-adjusted performance validates sentiment driven regime classification and volatility-aware adjustments.

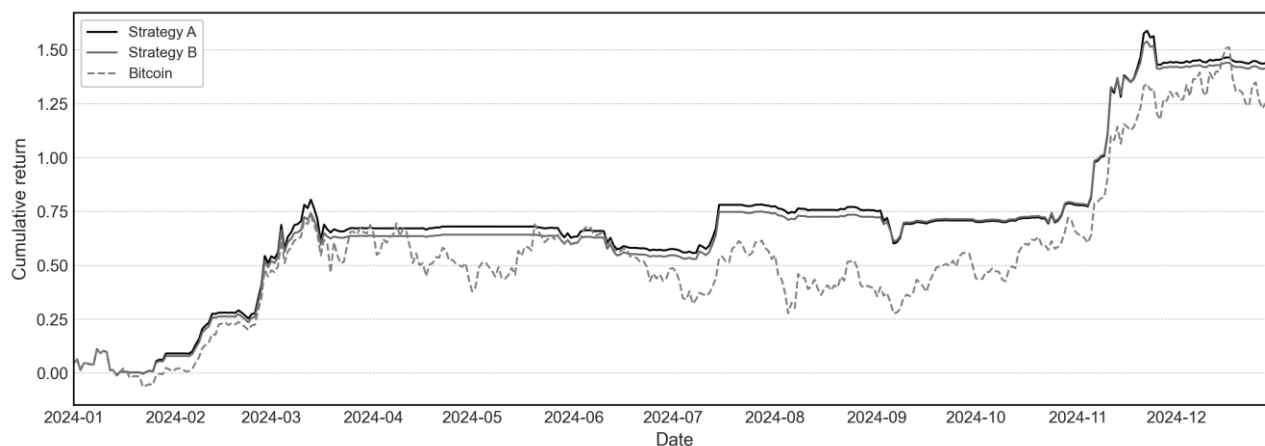


Figure 23: Cumulative Returns Strategy A&B (OOS)

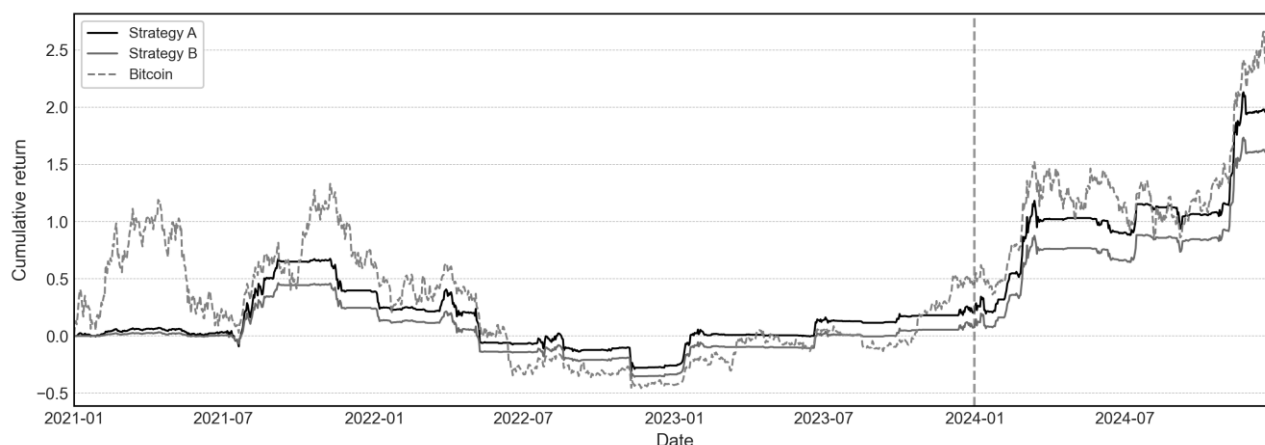


Figure 24: Cumulative Returns Strategy A&B (Full Period)

Figure 24 visualizes cumulative returns throughout the full evaluation period (2021–2024), clearly marking the IS/OOS split. The chart reveals three distinct phases in strategy performance.

1. **2021 Bull Market:** Strategies lag slightly as conservative allocation misses some upside
2. **2022 Bear Market:** Outperformance as strategies move to cash, avoiding 50%+ drawdowns
3. **2023-2024 Recovery:** Strategies re-enter at market lows and capture the subsequent rally

The divergence between strategies and Bitcoin buy-and-hold during 2022 represents the key value proposition: capital preservation during drawdowns enables compound growth advantages over full market cycles.

5.2.3 Extended Benchmark Comparison

Figure 25 and Table 16 provide an extended comparison against multiple benchmarks, including Bitcoin Buy-and-Hold, Dollar-Cost Averaging, the S&P 500, and the CCI30 Cryptocurrency Index.

Sample		Total Ret	Ann Vol	Sharpe	Max DD
OOS	Strategy A (comp)	1.43	0.33	2.7	0.14
	Strategy B (comp)	1.41	0.31	2.83	0.12
	Bitcoin	1.21	0.53	1.49	0.26
	DCA	1.12	0.53	1.41	0.26
	S&P500	0.23	0.13	1.65	0.08
	CCI30	0.78	0.58	1.0	0.4

Table 16: Performance Summary (OOS), extended

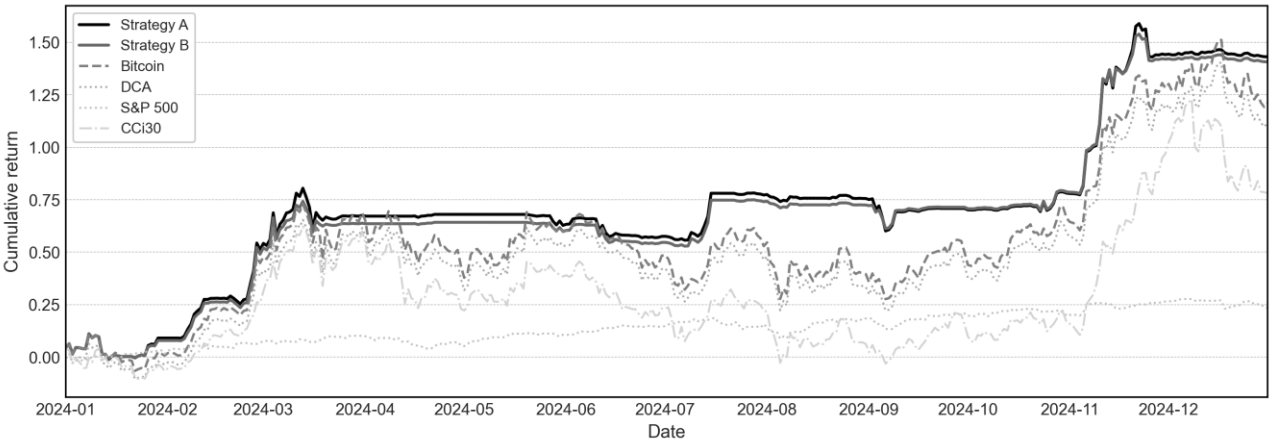


Figure 25: Extended Benchmark Comparison (2024)

Key observations include:

- Both sentiment driven strategies clearly outperform all other benchmarks, including Bitcoin buy-and-hold, particularly from a risk-adjusted perspective.
- Dollar-Cost Averaging significantly underperforms, highlighting that passive averaging strategies were not optimal during the strong market upturn of the OOS period.
- The S&P 500 and CCI30 demonstrate notably lower performance, reinforcing the strategic advantage of sentiment driven Bitcoin-specific approaches in capturing cryptocurrency market dynamics.

The strategies robust outperformance highlights their viability and superiority in both return generation and risk mitigation, validating sentiment based tactical allocation.

Summary of Empirical Findings:

- **Alpha generation:** Composite sentiment driven regime allocation consistently achieves higher risk-adjusted returns than passive benchmarks.
- **Composite effectiveness:** Clearly outperforms single-engine signals (VADER-only, FinBERT-only), confirming methodological diversity enhances robustness.
- **Volatility-aware adjustments:** Strategy B demonstrates considerable effectiveness in reducing drawdowns and volatility, affirming its value as a risk management enhancement.
- **These results directly address** the core research questions, providing empirical evidence for the feasibility and effectiveness of sentiment based quantitative Bitcoin trading strategies.

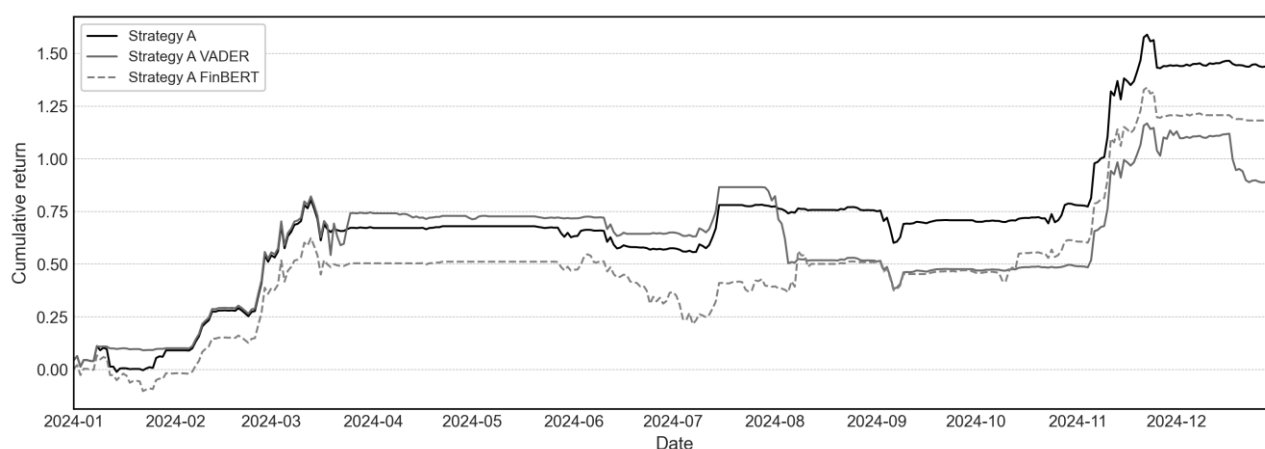


Figure 26: Single vs. Composite Sentiment Performance (2024)

Figure 26 visualizes these performance differences throughout 2024. The composite strategy consistently outperforms both individual approaches.

These findings show that the combined VADER-FinBERT approach represents the most effective sentiment analysis technique for Bitcoin trading, validating the hypothesis that methodological diversity enhances both returns and risk management in sentiment driven strategies.

5.3 Risk Analysis

This section examines the risk characteristics underlying the superior performance demonstrated in Section 5.2, revealing how sentiment driven strategies achieve their greater risk-adjusted returns.

5.3.1 Drawdown Analysis

Figure 27 illustrates the underwater periods for each strategy throughout the full evaluation period (2021–2024). Sentiment driven strategies demonstrate markedly lower and shorter drawdowns compared to the Bitcoin buy-and-hold strategy.

Key observations:

- Maximum drawdowns for sentiment driven strategies were significantly lower (Strategy A: 59%, Strategy B: 57%) compared to Bitcoin buy-and-hold (77%) during the severe market downturn of 2022.
- Strategies effectively mitigate downside risk, validating their protective characteristics and superior risk management capabilities.

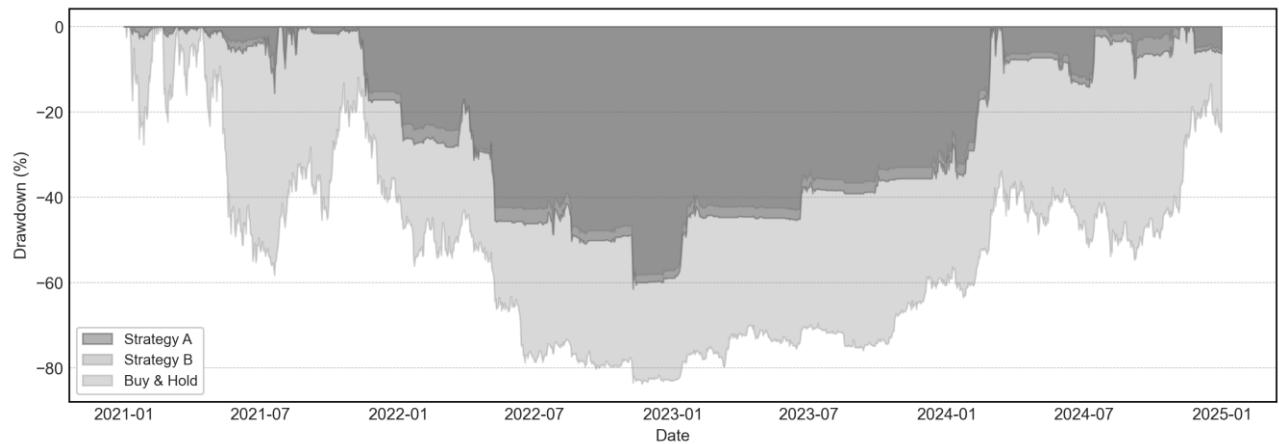


Figure 27: Underwater Plot (2021-2024)

5.3.2 Allocation Dynamics

Figure 28 visualizes the allocation adjustments for each strategy during the crucial out-of-sample period (2024). Dynamic allocation adjustments support the superior risk management, clearly showing tactical shifts in exposure based on sentiment driven regime classification.

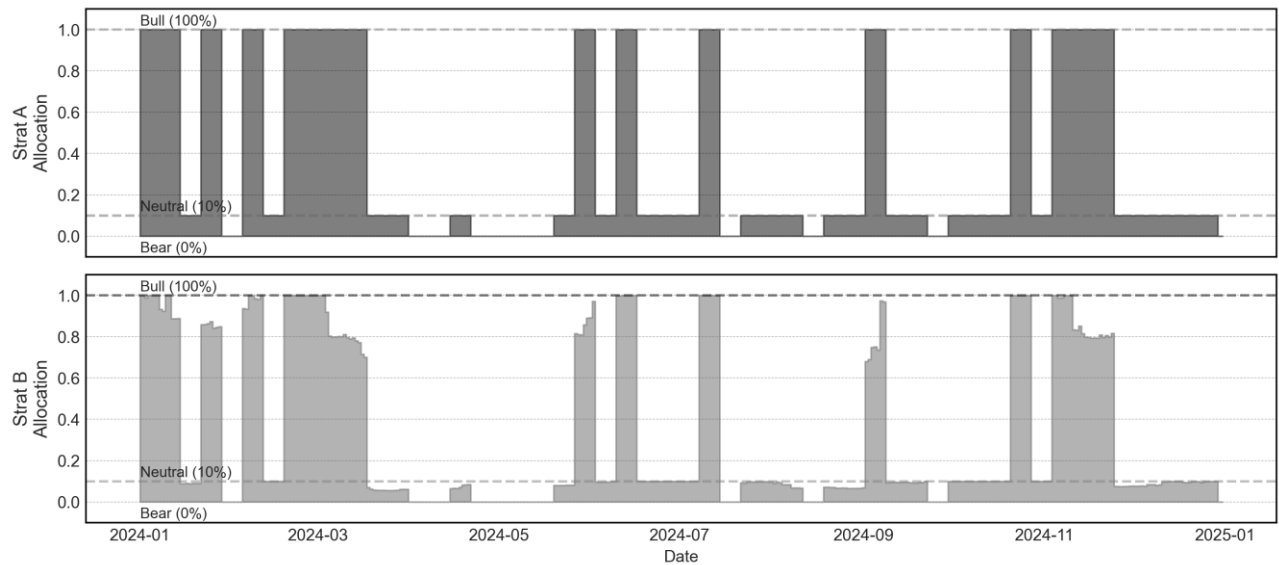


Figure 28: Allocation Map OOS

Table 17 summarize regime distribution statistics and average allocations for both strategies during the out-of-sample period, providing quantitative insights into their tactical allocation behaviour:

Regime Distribution	Strategy A (%)	Strategy B (%)
Bull (100% BTC)	25.39	25.12
Neutral (10% BTC)	49.35	42.57
Bear (0% BTC)	25.25	32.3
Average Allocation	30.33	25.88

Table 17: Regime Distribution Statistics

Key insights:

Strategy A: Exhibits clear binary regime shifts, creating distinct risk-on/risk-off periods:

- Bull regime (100% exposure): 25% of OOS period
- Neutral regime (10% exposure): approximately half of OOS period (49%)
- Bear regime (0% exposure): approximately 25% of OOS period

Strategy B: Volatility scaling creates smoother allocation transitions:

- Average allocation: Lower at 25.88%, indicating more conservative exposure overall compared to Strategy A's 30.33%
- More gradual position reductions during volatile periods
- Maximum allocation frequently constrained below 100% due to volatility-driven adjustments, effectively reducing exposure during heightened market uncertainty

Summary of Risk Management Effectiveness

The risk analysis conclusively demonstrates that both sentiment driven strategies, especially Strategy B, effectively control volatility and mitigate drawdown risk. Dynamic regime classification and volatility-aware adjustments ensure robust downside protection while still capturing significant upside potential, addressing critical investor concerns about Bitcoin's inherent market volatility.

5.3.3 Correlation Analysis

Understanding the correlation structure between sentiment driven strategies and Bitcoin provides crucial insights into diversification benefits and risk characteristics.

	Strategy A vs Bitcoin	Strategy B vs Bitcoin
Full Period	0.58	0.54
IS	0.55	0.51
OOS	0.68	0.67

Table 18: Strategy-Bitcoin Correlation Analysis

The correlation analysis reveals three important characteristics:

Moderate Correlation Levels: Both strategies maintain correlations between 0.54-0.68 with Bitcoin, indicating substantial but not complete dependence on underlying market movements. This moderate correlation suggests the strategies provide meaningful diversification benefits while remaining sufficiently exposed to capture Bitcoin's return potential.

Temporal Stability: The correlation remains relatively stable between in-sample (0.55) and out-of-sample (0.68) periods for Strategy A, with only a modest increase during the 2024 bull market. This stability suggests the strategy's risk characteristics are consistent across different market environments.

Strategy Similarity: The near-identical correlation profiles between Strategy A and Strategy B (difference < 0.01) confirm that volatility scaling preserves the fundamental relationship with Bitcoin while adjusting position sizes. This validates Strategy B as a risk-managed variant rather than a fundamentally different approach.

This correlation profile explains the superior risk-adjusted returns: the strategies capture approximately 65-70% of Bitcoin's movements while avoiding the full downside through tactical de-risking, creating an asymmetric return profile favourable to investors.

5.4 Robustness & Sensitivity Analysis

This section evaluates the robustness of the sentiment driven trading strategies by analysing their sensitivity to methodological choices and exploring the effectiveness of the composite sentiment signal compared to individual signals.

5.4.1 Comparative Analysis of Sentiment Signals

To assess whether the composite sentiment approach offers meaningful benefits over single sentiment indicators, Table 19 compares Strategy A's performance (OOS) when using:

- Composite sentiment (VADER + FinBERT)
- VADER sentiment only
- FinBERT sentiment only

Sample		Total Ret	Ann Vol	Sharpe	Max DD
OOS	Strategy A (comp)	1.43	0.33	2.7	0.14
	Strategy A VADER	0.88	0.37	1.71	0.26
	Strategy A FinBERT	1.18	0.38	2.03	0.25

Table 19: Single vs. Composite Sentiment Performance summary (OOS)

The composite strategy demonstrates clear superiority across all risk-adjusted metrics, notably:

- **Return Enhancement:** The composite approach achieves the highest total return (143%) compared to VADER-only (88%) and FinBERT-only (118%).
- **Risk-Adjusted Performance:** The Sharpe ratio comparison provides the most compelling evidence: the composite strategy (2.70) significantly outperforms both VADER (1.71) and FinBERT (2.03). This 58% improvement over VADER and 33% enhancement over FinBERT demonstrates that the combination captures complementary information from both methods.
- **Drawdown Protection:** The composite approach maintains competitive drawdown characteristics (14%) compared to individual methods, with VADER showing marginally better protection but at the cost of substantially lower returns.

5.4.2 Implementation Robustness

To ensure the strategy's performance is not dependent on specific implementation assumptions, Table 20 presents sensitivity analysis across different transaction cost levels and execution scenarios.

	Total Ret	Ann Vol	Sharpe	Max DD
Baseline (0.10% TC)	1.43	0.33	2.70	0.14
Low TC (0.05%)	1.45	0.33	2.73	0.14
High TC (0.20%)	1.39	0.33	2.65	0.14
1-day execution delay	1.37	0.34	2.53	0.18

Table 20: Implementation Robustness Tests - Strategy A (OOS)

The implementation robustness tests confirm the strategy's viability under realistic trading conditions:

Transaction Cost Resilience: Strategy A maintains strong performance across varying fee structures. Even with doubled transaction costs the Sharpe ratio remains exceptional at 2.65, representing only a 1.9% degradation from baseline. This minimal impact validates the weekly rebalancing frequency as optimal for balancing signal responsiveness with cost efficiency. The low cost scenario yields a marginal improvement confirming that performance is not heavily dependent on achieving institutional grade execution costs.

Execution Timing Robustness: The 1-day execution delay, simulating realistic order processing and settlement, results in a modest performance decline. The 6.3% Sharpe reduction and increased maximum drawdown (18% vs 14%) indicate that prompt execution provides value, though the strategy remains highly profitable

even with delayed implementation, which suggests the sentiment signals capture persistent regime characteristics.

Volatility Stability: Remarkably, annualized volatility remains constant across all scenarios, demonstrating that implementation variations do not materially affect the strategy's risk profile. This consistency ensures predictable portfolio behaviour regardless of execution quality.

All tested scenarios maintain Sharpe ratios above 2.5. This robustness confirms that Strategy A's outperformance stems from genuine sentiment driven alpha rather than fragile backtest assumptions.

These findings address practical deployment concerns, demonstrating that the sentiment driven approach remains viable for retail traders (facing higher costs) and institutional investors (potentially experiencing execution delays) alike.

5.5 Discussion of Results

The empirical findings presented in this chapter provide evidence for the viability and effectiveness of sentiment driven quantitative Bitcoin trading strategies. This section synthesizes the results in the context of the research questions and discusses their theoretical and practical implications.

RQ1: How can digital sentiment from various online sources be reliably extracted, quantified, and integrated into Bitcoin trading strategies?

The results demonstrate successful sentiment extraction and integration through a multi stage pipeline. The composite approach combining VADER and FinBERT sentiment scores, normalized through 90-day rolling z-scores and smoothed with 5-period EMA, creates actionable trading signals. The regime classification system (75th/25th percentiles) effectively translates continuous sentiment into discrete trading decisions, achieving out-of-sample Sharpe ratios of 2.70, significantly exceeding the Bitcoin buy-and-hold benchmark (1.49).

RQ2: Which sentiment analysis techniques are most effective for creating sentiment feeds to inform Bitcoin trading decisions?

The comparative analysis in Section 5.4.1 definitively answers this question: the composite VADER-FinBERT approach outperforms individual methods by 33-58% in risk-adjusted terms. This superiority stems from methodological complementarity, VADER captures social media expressions while FinBERT understands financial context. The combination reduces false signals and enhances regime identification accuracy, validating the hypothesis that methodological diversity improves trading outcomes.

RQ3: How does the inclusion of sentiment data influence alpha generation and overall performance of Bitcoin trading strategies?

The inclusion of sentiment data fundamentally changed the investment outcome. The strategy captured the majority of the upside during the 2024 bull market 143% return vs. Bitcoin's 121%, while simultaneously providing crucial downside protection by halving the maximum drawdown (14% vs. 26%). This demonstrates that the primary value of the sentiment signal was not necessarily in forecasting future returns, a task at which its predictive power was found to be limited, but rather in accurately identifying the current market regime. This allowed the strategy to effectively manage risk by de-risking during periods of negative sentiment, thereby preserving capital and achieving superior performance over a full market cycle.

Theoretical Implications

The results support behavioural finance theories in cryptocurrency markets. The success of sentiment driven strategies validate that Bitcoin markets exhibit exploitable inefficiencies driven by investor emotions, contradicting with strong form EMH. The correlation analysis (0.54-0.68) reveals that sentiment strategies capture market regime changes while maintaining sufficient independence to add value, aligning with literature on sentiment's role in cryptocurrency markets (Mai et al., 2018; Kraaijeveld & De Smedt, 2020).

Key Success Factors

The analysis reveals critical elements driving strategy performance:

- Composite signals providing methodological diversity through VADER-FinBERT combination
- Adaptive thresholds adjusting to evolving sentiment distributions via rolling percentiles
- Weekly rebalancing balancing signal responsiveness with transaction costs
- Risk-first approach prioritizing Sharpe ratio optimization over absolute returns

Conclusion of Results

The results confirm that digital sentiment, when properly quantified and integrated, represents a valuable factor for tactical asset allocation in Bitcoin. The strategy's success is rooted in its ability to systematically respond to the collective mood of the market, sidestepping major downturns while participating in upward trends, a clear validation of a behaviourally driven investment approach in cryptocurrency markets.

6 Conclusion & Future Work

6.1 Summary of Findings

In this thesis an exploratory sentiment driven Bitcoin trading prototype was formulated, implemented, and evaluated. Public Reddit data was processed by a lexicon-based method (VADER) and a transformer-based model (FinBERT). After z-normalisation and exponential smoothing, the two signals were combined into a composite indicator that was used to classify market regimes. A weekly, un-levered rebalancing rule was then applied. Out-of-sample results for 2024 indicated that the composite strategy achieved a total return of 143%, an annualised Sharpe ratio of 2.70, and a maximum draw-down of 14%, thereby outperforming Bitcoin buy-and-hold, dollar-cost averaging, the S&P 500, and the CCI30 index. A volatility aware variant further reduced draw-downs with only marginal sacrifice in return. Robustness tests confirmed the strategy's resilience to varying transaction costs and execution delays, with Sharpe ratios remaining above 2.5 across all scenarios. Consequently, the three research questions were answered positively: reliable extraction and aggregation of digital sentiment were demonstrated; the composite VADER-FinBERT feed proved to be the most effective of the tested alternatives; and the inclusion of sentiment was shown to improve both absolute and risk-adjusted performance.

Theoretical Contributions

The thesis provided empirical evidence that information inefficiencies persist in Bitcoin markets, thereby challenging the weak-form Efficient Market Hypothesis. The documented correlation between retail sentiment and contemporaneous returns supported key assumptions of behavioural finance regarding the influence of investor emotion and herd behaviour in speculative environments. A methodological contribution was delivered by illustrating how heterogeneous natural language processing tools can be combined to offset individual weaknesses while still retaining interpretability.

Practical Contributions

A transparent, resource efficient pipeline was supplied that may be replicated without proprietary data or high-performance computing resources. The prototype showed that sentiment filters and volatility caps can approximately halve draw-downs relative to passive exposure while remaining profitable under retail-level fee structures. The use of open-source tools (VADER, FinBERT) and publicly available data democratizes quantitative trading, enabling individual investors to implement institutional grade strategies without significant capital or infrastructure requirements. These characteristics make the approach attractive for market participants who seek Bitcoin exposure but remain concerned about downside risk.

Limitations

Several limitations must be acknowledged. First, the analysis timeframe was restricted to the 2021-2024 period, so structural breaks beyond this horizon were not captured. Second, only Reddit discourse was considered, whereas institutional grade news sentiment, X (Twitter) sentiment, and on-chain indicators were excluded. Furthermore, the strategy's reliance on 60- and 90-day rolling windows for normalisation and signal generation means it remains inactive during the initial three months of any given period, a methodological trade-off for creating adaptive signals that avoid look-ahead bias. Third, execution assumptions, such as zero slippage and fills at the Monday close, simplified real-world trading frictions. Finally, the prototype was evaluated solely in backtests; live deployment and potential feedback effects on market behaviour were not examined.

6.2 Future Work

The promising results of this thesis open up several opportunities for future research. A crucial next step would be to move from historical simulation to practice by developing and deploying the strategy in a live environment. This would involve building a robust data pipeline to study real-world challenges like order book slippage and API latency. The framework could also be significantly enhanced by incorporating a richer set of data sources, such as on-chain metrics and professional news feeds, to create a more holistic sentiment signal. Furthermore, the methodology could be generalized beyond Bitcoin, by testing its applicability across other cryptocurrencies to determine if these sentiment dynamics are a feature of the broader digital asset class, and then potentially extending it to traditional assets to see if similar behavioural patterns can be exploited. Finally, more advanced methods like reinforcement learning might be explored to create a fully dynamic allocation agent, moving beyond fixed rules to a model that learns and adapts its strategy over time.

6.3 Final Remarks

By integrating publicly available sentiment data into a controlled allocation framework, the thesis has demonstrated that meaningful excess returns and substantial risk reduction can be realised in Bitcoin trading. Although the study remains exploratory and subject to the stated limitations, its findings suggest that digital sentiment represents a valuable and actionable information set. It is therefore concluded that sentiment analysis should occupy a central role in future quantitative crypto research and practice.

References

- Ahmed, S., Grobys, K., & Sapkota, N. (2020). Profitability of technical trading rules among cryptocurrencies with privacy function. *Finance Research Letters*, 35, 101495. <https://doi.org/10.1016/j.frl.2020.101495>
- Alajbeg, D., Bubas, Z., & Sonje, V. (2012). The efficient market hypothesis: Problems with interpretations of empirical tests. *Financial Theory and Practice*, 36(1), 53–72. <https://doi.org/10.3326/fintp.36.1.3>
- Almeida, J., & Gonçalves, T. C. (2023). A systematic literature review of investor behavior in the cryptocurrency markets. *Journal of Behavioral and Experimental Finance*, 37, 100785. <https://doi.org/10.1016/j.jbef.2022.100785>
- API, E. H. P. and F. D. (n.d.). *The Best API for Historical Stock Market Prices and Fundamental Financial Data* |Free Trial API. Retrieved July 26, 2025, from <https://eodhd.com>
- Araci, D. (2019). *FinBERT: Financial Sentiment Analysis with Pre-trained Language Models*. <https://doi.org/10.48550/arXiv.1908.10063>
- Barberis, N., & Thaler, R. (2002). *NBER WORKING PAPER SERIES*.
- Baur, D. G., Hong, K., & Lee, A. D. (2018). Bitcoin: Medium of exchange or speculative assets? *Journal of International Financial Markets, Institutions and Money*, 54, 177–189. <https://doi.org/10.1016/j.intfin.2017.12.004>
- Bikhchandani, S., & Sharma, S. (2000). Herd Behavior in Financial Markets. *IMF Staff Papers*, 47(3), 279–310. <https://doi.org/10.2307/3867650>
- Bollen, J., Mao, H., & Zeng, X. (2011). Twitter mood predicts the stock market. *Journal of Computational Science*, 2(1), 1–8. <https://doi.org/10.1016/j.jocs.2010.12.007>
- Brandvold, M., Molnar, P., Vagstad, K., & Valstad, O. C. A. (2015). *Price Discovery on Bitcoin Exchanges*. *Coinmarketcap*. (n.d.). CoinMarketCap. Retrieved April 6, 2025, from <https://coinmarketcap.com/currencies/bitcoin/>
- Corbet, S., Lucey, B., Urquhart, A., & Yarovaya, L. (2019). Cryptocurrencies as a financial asset: A systematic analysis. *International Review of Financial Analysis*, 62, 182–199. <https://doi.org/10.1016/j.irfa.2018.09.003>
- Crypto Research, Data, Tools—Explore Behavioral Analytics*. (n.d.). Santiment. Retrieved July 26, 2025, from <https://santiment.net/>
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*.
- Ernest P. Chan. (2021). *Quantitative Trading: How to Build Your Own Algorithmic Trading Business* (2nd ed.). Wiley.

- Esuli, A., & Sebastiani, F. (2006). *SentiWordNet: A High-Coverage Lexical Resource for Opinion Mining*.
- Fama, E. F. (1970). *Efficient Capital Markets: A Review of Theory and Empirical Work*.
- Fang, F., Ventre, C., Basios, M., Kanthan, L., Martinez-Rego, D., Wu, F., & Li, L. (2022). Cryptocurrency trading: A comprehensive survey. *Financial Innovation*, 8(1), 13. <https://doi.org/10.1186/s40854-021-00321-6>
- Fischer, T., & Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2), 654–669. <https://doi.org/10.1016/j.ejor.2017.11.054>
- Garcia, D., & Schweitzer, F. (2015). Social signals and algorithmic trading of Bitcoin. *Royal Society Open Science*, 2(9), 150288. <https://doi.org/10.1098/rsos.150288>
- Glaser, Fl. (2014). BITCOIN - ASSET OR CURRENCY? REVEALING USERS' HIDDEN INTENTIONS. *Tel Aviv*.
- Goodell, J. W., & Goutte, S. (2021). Co-movement of COVID-19 and Bitcoin: Evidence from wavelet coherence analysis. *Finance Research Letters*, 38, 101625. <https://doi.org/10.1016/j.frl.2020.101625>
- Hudson, R., & Urquhart, A. (2021). Technical trading and cryptocurrencies. *Annals of Operations Research*, 297(1–2), 191–220. <https://doi.org/10.1007/s10479-019-03357-1>
- Hurjui, R. B. (2022). *The Effect of Market Sentiment on Cryptocurrency Prices*.
- Hutto, C., & Gilbert, E. (2014). VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1), 216–225. <https://doi.org/10.1609/icwsm.v8i1.14550>
- Jiang, Z., Xu, D., & Liang, J. (2017). *A Deep Reinforcement Learning Framework for the Financial Portfolio Management Problem* (arXiv:1706.10059). arXiv. <https://doi.org/10.48550/arXiv.1706.10059>
- Kahneman, D., & Tversky, A. (1979). *Prospect Theory: An Analysis of Decision under Risk*.
- Kraaijeveld, O., & De Smedt, J. (2020). The predictive power of public Twitter sentiment for forecasting cryptocurrency prices. *Journal of International Financial Markets, Institutions and Money*, 65, 101188. <https://doi.org/10.1016/j.intfin.2020.101188>
- Kristoufek, L. (2013). BitCoin meets Google Trends and Wikipedia: Quantifying the relationship between phenomena of the Internet era. *Scientific Reports*, 3, 3415. <https://doi.org/10.1038/srep03415>
- Kyriazis, N., Papadamou, S., & Corbet, S. (2020). A systematic review of the bubble dynamics of cryptocurrency prices. *Research in International Business and Finance*, 54, 101254. <https://doi.org/10.1016/j.ribaf.2020.101254>

- Liu, B. (2012). Sentiment Analysis and Opinion Mining. In *Synthesis Lectures on Human Language Technologies* (Vol. 5). <https://doi.org/10.2200/S00416ED1V01Y201204HLT016>
- Mai, F., Shan, Z., Bai, Q., Wang, X. (Shane), & Chiang, R. H. L. (2018). How Does Social Media Impact Bitcoin Value? A Test of the Silent Majority Hypothesis. *Journal of Management Information Systems*, 35(1), 19–52. <https://doi.org/10.1080/07421222.2018.1440774>
- Makarov, I., & Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2), 293–319. <https://doi.org/10.1016/j.jfineco.2019.07.001>
- Mehta, I. (2023, March 30). Twitter announces new API with only free, basic and enterprise levels. *TechCrunch*. <https://techcrunch.com/2023/03/29/twitter-announces-new-api-with-only-free-basic-and-enterprise-levels/>
- Milevsky, M., & Eposner, S. (2011). A continuous-time reexamination of dollar-cost averaging. *International Journal of Theoretical and Applied Finance*, 06. <https://doi.org/10.1142/S0219024903001888>
- Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*.
- Otabek, S., & Choi, J. (2024). Multi-level deep Q-networks for Bitcoin trading strategies. *Scientific Reports*, 14(1). <https://doi.org/10.1038/s41598-024-51408-w>
- Petukhina, A., Trimborn, S., Härdle, W. K., & Elendner, H. (2018). Investing With Cryptocurrencies – Evaluating the Potential of Portfolio Allocation Strategies. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.3274193>
- Phillips, R. C., & Gorse, D. (2017). Predicting cryptocurrency price bubbles using social media data and epidemic modelling. *2017 IEEE Symposium Series on Computational Intelligence (SSCI)*, 1–7. <https://doi.org/10.1109/SSCI.2017.8280809>
- Ravi, K., & Ravi, V. (2015). A survey on opinion mining and sentiment analysis: Tasks, approaches and applications. *Knowledge-Based Systems*, 89, 14–46. <https://doi.org/10.1016/j.knosys.2015.06.015>
- Reuters. (2022, November 12). Crypto markets in turmoil over FTX bankruptcy. *Reuters*. <https://www.reuters.com/technology/crypto-markets-turmoil-over-ftx-bankruptcy-2022-11-11/>
- Rivin, I., & Scevola, C. (2018). *The CCI30 Index* (arXiv:1804.06711). arXiv. <https://doi.org/10.48550/arXiv.1804.06711>
- Rob J Hyndman, George Athanasopoulos. (2021). *Forecasting: Principles and Practice (3rd ed)*. <https://otexts.com/fpp3/>

Shen, S., & Galbraith, A. (2021, June 25). China's ban forces some bitcoin miners to flee overseas, others sell out. *Reuters*. <https://www.reuters.com/technology/chinas-ban-forces-some-bitcoin-miners-flee-overseas-others-sell-out-2021-06-25/>

Shiller, R. J. (2003). From Efficient Markets Theory to Behavioral Finance. *The Journal of Economic Perspectives*, 17(1), 83–104. JSTOR.

Taboada, M., Brooke, J., Tofiloski, M., Voll, K., & Stede, M. (2011). Lexicon-Based Methods for Sentiment Analysis. *Computational Linguistics*, 37(2), 267–307. https://doi.org/10.1162/coli_a_00049

Technology and insights for data-driven companies. (n.d.). RavenPack. Retrieved July 26, 2025, from <https://www.ravenpack.com>

Tetlock, P. C. (2007). Giving Content to Investor Sentiment: The Role of Media in the Stock Market. *The Journal of Finance*, 62(3), 1139–1168. <https://doi.org/10.1111/j.1540-6261.2007.01232.x>

Troy Segal. (2024, June 21). *Fundamental Analysis: Principles, Types, and How to Use It*. Investopedia. <https://www.investopedia.com/terms/f/fundamentalanalysis.asp>

Vidal-Tomás, D. (2021). The entry and exit dynamics of the cryptocurrency market. *Research in International Business and Finance*, 58, 101504. <https://doi.org/10.1016/j.ribaf.2021.101504>

Wiggers, K. (2023, April 18). Reddit will begin charging for access to its API. *TechCrunch*. <https://techcrunch.com/2023/04/18/reddit-will-begin-charging-for-access-to-its-api/>

Xu, Q. A., Chang, V., & Jayne, C. (2022). A systematic review of social media-based sentiment analysis: Emerging trends and challenges. *Decision Analytics Journal*, 3, 100073. <https://doi.org/10.1016/j.da-jour.2022.100073>

Yermack, D. (2014). *Is bitcoin a real currency? An economic appraisal*.

Zhang, L., Wang, S., & Liu, B. (2018). Deep learning for sentiment analysis: A survey. *WIREs Data Mining and Knowledge Discovery*, 8(4). <https://doi.org/10.1002/widm.1253>

List of aids

Aid	Usage	Affected areas
Perplexity Pro, Claude 3.7	Confirming that it fits to what was written already in “2. Literature review” & confirming clarity, help define delimitations to align with RQs and feasibility	Chapter 1
Perplexity Pro, GPT 4.5, Claude 3.7	Extended browsing for relevant literature, additional source to Google Scholar and Research Gate. Used to refine and give advice for a good structure framework for the literature review Improvement of sentence structures and flow Extended research of strategies, blueprinting the structure enhancement of sentences	Chapter 2 & Chapter 3
Gemini 2.5	Debugging for VADER files, also mentioned in the comments of the code	Prototype, Data Processing, VADER
Gemini 2.5	Help script the Collab Notebook for FinBERT	Prototype, Data Processing FinBERT
GPT 4o	Bug fixing of Google Trends data script	Prototype, Google Trends Data
Gemini 2.5, Claude 3.7	Bug fixing of strategy implementation scripts, exploration of possible enhancements and robustness tests	Prototype strategies
GPT 4o, GPT o3, GPT 4.5	Grammar and interpretation check Debugging and structuring of strategies Exploration of strategy improvements	Chapter 4
GPT o3, Gemini 2.5 Pro Claude Opus 4	Confirming that interpretations made, make actually sense and are correctly expressed Debugging & Improvement of code for the Benchmark analysis Confirming that chosen metrics and functions are correct	Chapter 5
Claude Opus 4, GPT 4.5, Gemini 2.5 Pro	“AI Supervisor” screening of chapter structure and enhancement suggestions	Chapter 6

List of Figures

Figure 1: Bitcoin Daily Closing Prices (until March 2025)	1
Figure 2: Iterative modelling workflow for exploratory sentiment analysis and quantitative strategy development, inspired by Hyndman & Athanasopoulos (2021)	17
Figure 3: Python workflow to generate VADER sentiment scores	25
Figure 4: FinBERT Python setup	27
Figure 5: BTC Daily Closing Prices 2021 – 2024	29
Figure 6: Histogram of daily log-returns (2021-2024)	29
Figure 7: Rolling Volatility 2021-2024	30
Figure 8: Histogram of daily aggregated VADER sentiment	30
Figure 9: ACF: VADER-scaled Sentiment (lags 1-30)	31
Figure 10: Histogram of daily aggregated FinBERT sentiment	32
Figure 11: ACF: FinBERT-scaled Sentiment (lags 1-30)	32
Figure 12: Net Weekly Google Trends Sentiment	33
Figure 13: ACF: Weekly Net Google-Trends Sentiment (lags 1–30)	33
Figure 14: News Sentiment Plot	34
Figure 15: Weekly BTC log-returns against delta VADER Sentiment	35
Figure 16: VADER Δ sentiment vs BTC returns, lag–lead correlation	36
Figure 17: Weekly BTC log-returns against delta FinBERT Sentiment	37
Figure 18: FinBERT Δ sentiment vs BTC returns, lag–lead correlation	37
Figure 19: Weekly BTC log-returns against delta Google Sentiment	39
Figure 20: Google Trends Δ -sentiment vs BTC returns, Lag–Lead Correlation Map	39
Figure 21: Weekly BTC log-returns against Δ Combined Sentiment (z-scaled average)	40
Figure 22: Combined Δ -sentiment vs BTC returns, Lag–Lead Correlation Map	41
Figure 23: Cumulative Returns Strategy A&B (OOS)	51
Figure 24: Cumulative Returns Strategy A&B (Full Period)	51
Figure 25: Extended Benchmark Comparison (2024)	52

Figure 26: Single vs. Composite Sentiment Performance (2024)	53
Figure 27: Underwater Plot (2021-2024)	54
Figure 28: Allocation Map OOS	54

List of Tables

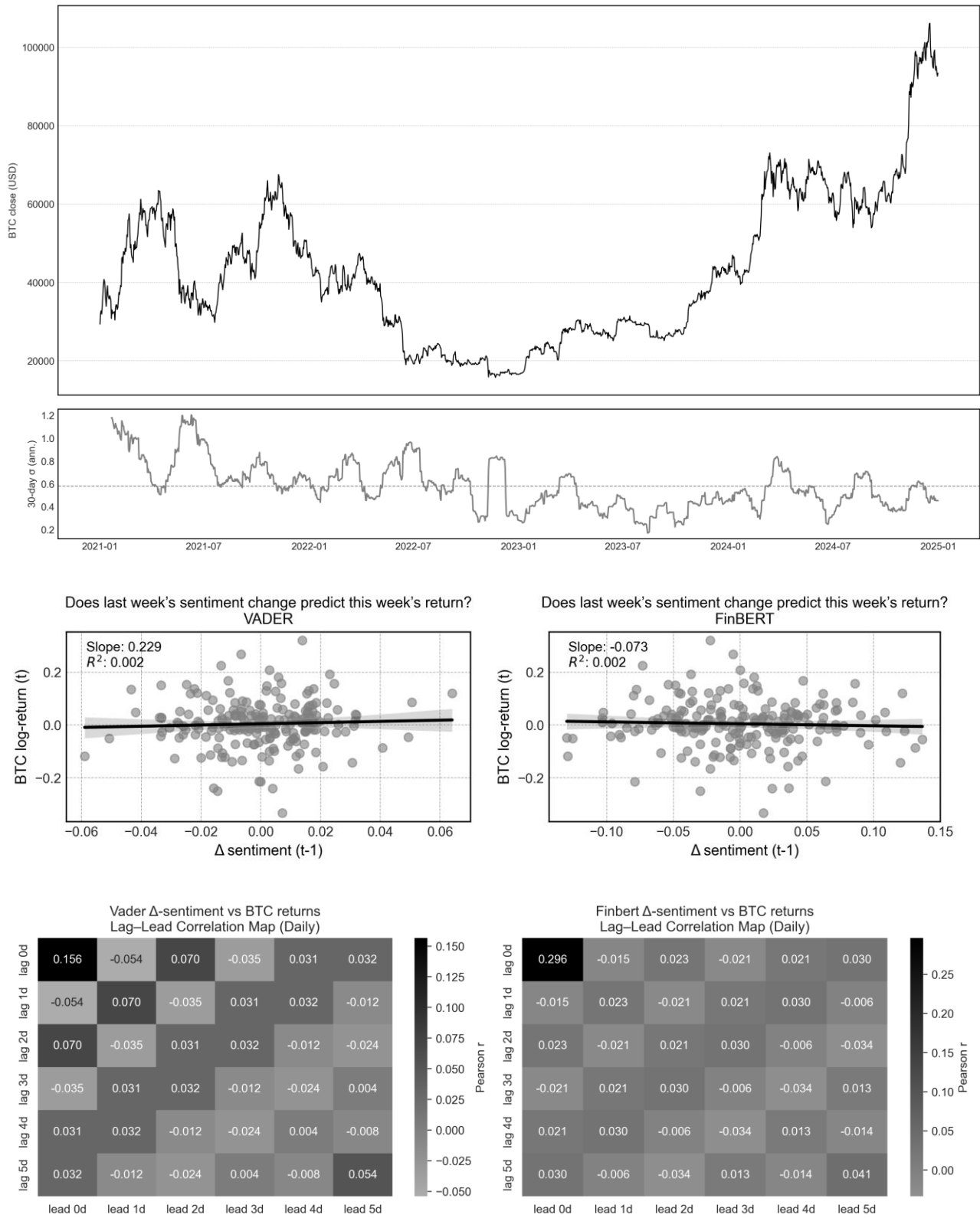
Table 1: Sentiment Analysis Tools comparison	18
Table 2: Volume of the raw reddit data	22
Table 3: Search Terms for Bitcoin Sentiment	23
Table 4: Overview of Reddit Data Before and After initial Preprocessing	24
Table 5: Example of Reddit Data with VADER Sentiment Scores	25
Table 6: Example of Reddit Data with FinBERT Sentiment Scores	27
Table 7: Summary statistics of daily log-returns (2021 - 2024)	29
Table 8: Summary statistics of scaled VADER sentiment	31
Table 9: Summary statistics of scaled FinBERT sentiment	32
Table 10: Findings of Chapter 4.2	42
Table 11: Signal Engineering	43
Table 12: Hyperparameters	44
Table 13: Neutral Allocation & EMA, Grid Search Results	49
Table 14: Performance Summary (IS)	50
Table 15: Performance Summary (OOS)	50
Table 16: Performance Summary (OOS), extended	52
Table 17: Regime Distribution Statistics	55
Table 18: Strategy-Bitcoin Correlation Analysis	56
Table 19: Single vs. Composite Sentiment Performance summary (OOS)	57
Table 20: Implementation Robustness Tests - Strategy A (OOS)	57

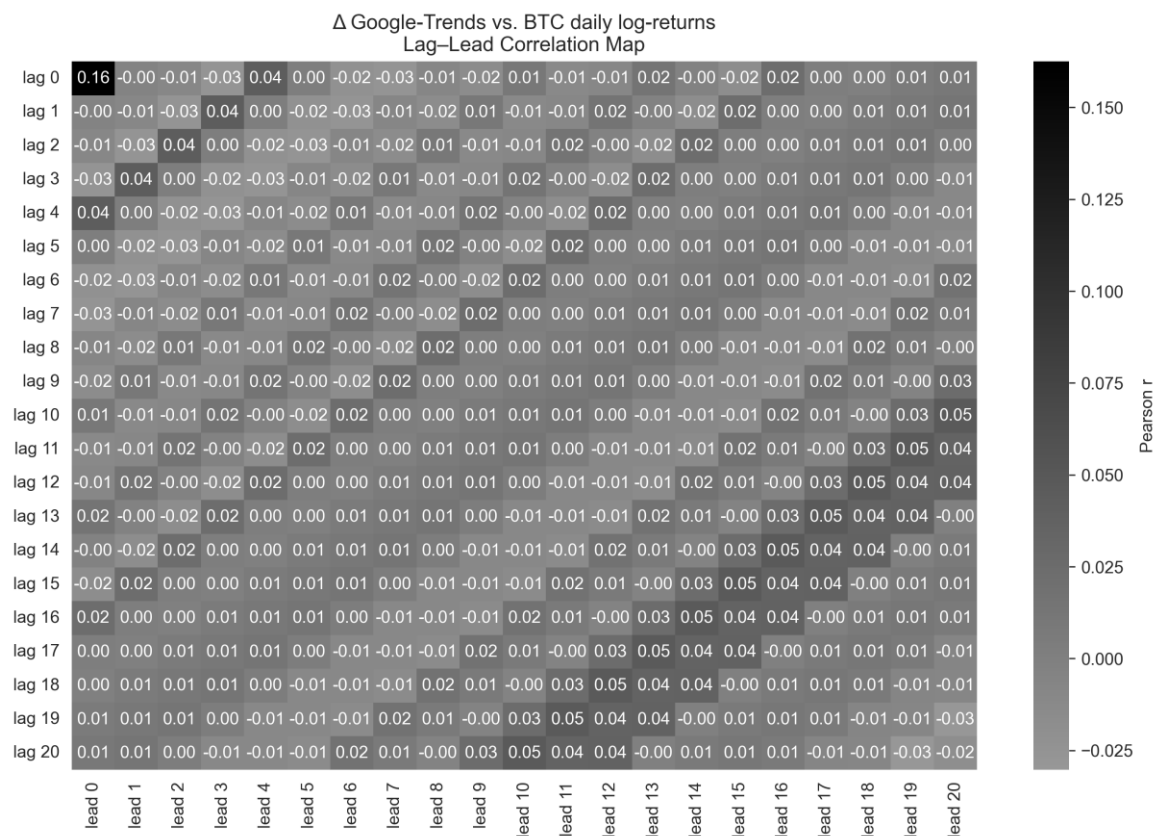
List of Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
BTC	Bitcoin (ticker symbol)
CCi30	Cryptocurrency Index of the 30 largest coins
DCA	Dollar-Cost Averaging
DD	Draw-Down (used in “max DD”)
EMA	Exponential Moving Average
EMH	Efficient-Market Hypothesis
FinBERT	Financial-domain Bidirectional Encoder Representations from Transformers
GPU	Graphics Processing Unit
IS / OOS	In-Sample / Out-of-Sample
JSON	JavaScript Object Notation (data-exchange format)
MA	Moving Average
ML	Machine Learning
NLP	Natural Language Processing
P&L	Profit and Loss
RQ	Research Question
VADER	Valence Aware Dictionary and Sentiment Reasoner

Appendix

A: EDA Sentiment, additional insights





B: Software and Libraries List

Python	v3.11	NLTK	v3.8.1
Pandas	v2.2.3	Transformers	v4.51.3
NumPy	v2.3.2	PyTorch	v2.7.0

C: Github Codebase & Data Link

Github Repository (public)

<https://github.com/leohubm/thesis>

Original Data

<https://drive.switch.ch/index.php/s/BGux9YPKsiyjOIG>

Password: THESIS2025LH

Expiration: 31-10-2025